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CyberOps Associate (Version 1.0) - CyberOps Associate Course Final Exam Answers

1. Which two statements are characteristics of a virus? (Choose two.)

- A virus typically requires end-user activation.
- A virus can be dormant and then activate at a specific time or date.
- A virus replicates itself by independently exploiting vulnerabilities in networks.
- A virus has an enabling vulnerability, a propagation mechanism, and a payload.
- A virus provides the attacker with sensitive data, such as passwords

Explanation: Topic 14.1.2

The type of end user interaction required to launch a virus is typically opening an application, opening a web page, or powering on the computer. Once activated, a virus may infect other files located on the computer or other computers on the same network.

2. What is a characteristic of a Trojan horse as it relates to network security?

- Too much information is destined for a particular memory block, causing additional memory areas to be affected

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network device interface.

- An electronic dictionary is used to obtain a password to be used to infiltrate a key network device.
- Malware is contained in a seemingly legitimate executable program.

Explanation: Topic 14.1.3

A Trojan horse carries out malicious operations under the guise of a legitimate program. Denial of service attacks send extreme quantities of data to a particular host or network device interface. Password attacks use electronic dictionaries in an attempt to learn passwords. Buffer overflow attacks exploit memory buffers by sending too much information to a host to render the system inoperable.

3. What technique is used in social engineering attacks?

- sending junk email
- buffer overflow
- phishing
- man-in-the-middle

Explanation: Topic 14.2.6

A threat actor sends fraudulent email which is disguised as being from a legitimate, trusted source to trick the recipient into installing malware on their device, or to share personal or financial information.

4. What is a purpose of implementing VLANs on a network?

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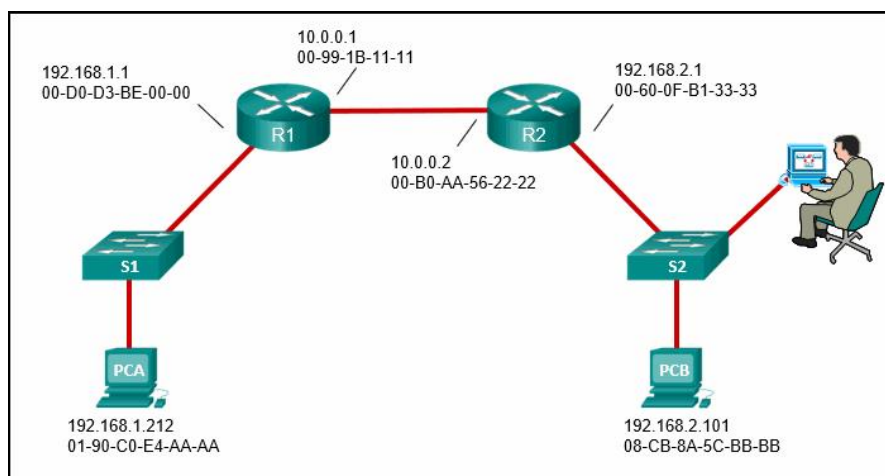


- They can separate user traffic.
- They prevent Layer 2 loops.
- They eliminate network collisions.
- They allow switches to forward Layer 3 packets without a router.

Explanation: Topic 11.1.12

VLANs are used on a network to separate user traffic based on factors such as function, project team, or application, without regard for the physical location of the user or device.

5. Refer to the exhibit. A cybersecurity analyst is viewing packets forwarded by switch S2. What addresses will identify frames containing data sent from PCA to PCB?



Src MAC: 00-60-0F-B1-33-33

Dst IP: 192.168.2.101

Dst MAC: 08-CB-8A-5C-BB-BB

- Src IP: 192.168.1.212

Src MAC: 01-90-C0-E4-AA-AA

Dst IP: 192.168.2.101

Dst MAC: 08-CB-8A-5C-BB-BB

- Src IP: 192.168.1.212

Src MAC: 00-60-0F-B1-33-33

Dst IP: 192.168.2.101

Dst MAC: 08-CB-8A-5C-BB-BB

- Src IP: 192.168.1.212

Src MAC: 00-60-0F-B1-33-33

Dst IP: 192.168.2.101

Dst MAC: 00-D0-D3-BE-00-00

Explanation: Topic 8.1.2

When a message sent from PCA to PCB reaches router R2, some frame header fields will be rewritten by R2 before forwarding to switch S2. The frames will contain the source MAC address of router R2 and the destination MAC address of PCB. The frames will retain the original IPv4 addressing applied by PCA which is the IPv4 address of PCA as the source address and the IPv4 address of PCB as the destination.

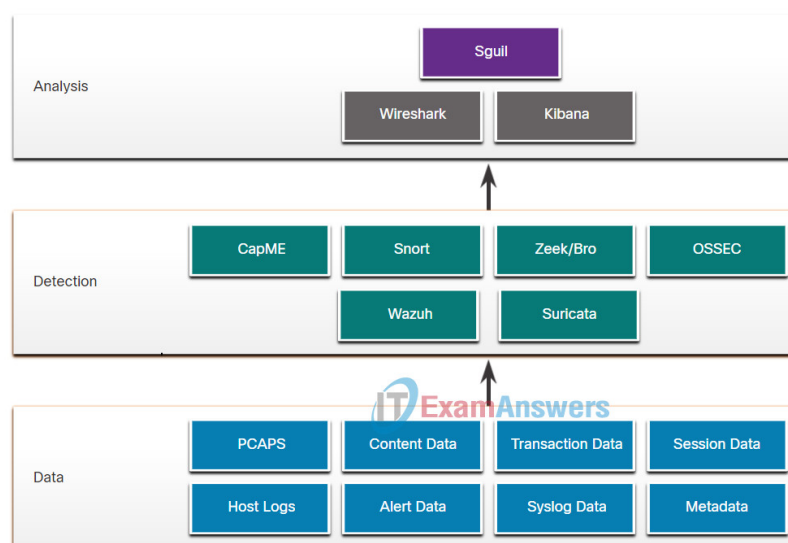
6. A cybersecurity analyst needs to collect alert data. What are three detection tools to perform this task in the Security Onion architecture? (Choose three.)

- CapME
- Wazuh
- Kibana
- Zeek
- Sguil
- Wireshark



Explanation: Topic 26.1.2

A Security Onion Architecture:



Detection tools in Security Onion Architecture include:
CapME, Snort, Zeek, OSSEC, Wazuh, Suricata.

7. Match the Security Onion tool with the description.

Match the Security Onion tool with the description.

Snort	network-based intrusion detection system
OSSEC	
Sguil	packet capture application
Wireshark	host-based intrusion detection system
	high-level cybersecurity analysis console

Snort

Wireshark

OSSEC

Sguil

Explanation: Topic 26.1.2

network-based intrusion detection system → Snort

packet capture application → Wireshark

host-based intrusion detection system → OSSEC

high-level cybersecurity analysis console → Sguil

8. In network security assessments, which type of test is used to evaluate the risk posed by vulnerabilities to a specific organization including assessment of the likelihood of attacks and the impact of successful exploits on the organization?

- port scanning
- risk analysis
- penetration testing
- vulnerability assessment

Explanation: Topic 23.1.4

A risk analysis includes assessment of the likelihood of attacks, identifies types of likely threat actors, and evaluates the impact of successful exploits on the organization.



(Not all options are used.)

user accounts	the parameters defining user access and behavior
listening ports	
service accounts	user accounts
software environment	the number of times the server is powered on and off
	the TCP and UDP daemons and ports that are allowed to be open on the server
	listening ports
	the tasks, processes, and applications that are permitted to run on the server
	software environment
	the definitions of the type of service that an application is allowed to run on a given host
	service accounts

Explanation: Topic 23.1.2

The elements of a server profile include the following:

- Listening ports – the TCP and UDP daemons and ports that are allowed to be open on the server
- User accounts – the parameters defining user access and



an application is allowed to run on a given host
Software environment – the tasks, processes, and applications that are permitted to run on the server

10. In addressing an identified risk, which strategy aims to shift some of the risk to other parties?

- risk avoidance
- risk sharing
- risk retention
- risk reduction

Explanation: Topic 23.3.1

There are four potential strategies for responding to risks that have been identified:

Risk avoidance – Stop performing the activities that create risk.

Risk reduction – Decrease the risk by taking measures to reduce vulnerability.

Risk sharing – Shift some of the risk to other parties.

Risk retention – Accept the risk and its consequences.

11. What is a network tap?

- a technology used to provide real-time reporting and long-term analysis of security events
- a Cisco technology that provides statistics on packets flowing through a router or multilayer switch
- a feature supported on Cisco switches that enables the switch to copy frames and forward them to an analysis device
- a passive device that forwards all traffic and physical layer errors to an analysis device

Explanation: Topic 15.1.3

A network tap is used to capture traffic for monitoring the network. The tap is typically a passive splitting device



including physical layer errors, to an analysis device.

12. Match the monitoring tool to the definition.

NetFlow	presents real-time reporting and long-term analysis of security events
Wireshark	
SNMP	
SIEM	

SIEM

provides statistics on packets flowing through a Cisco router or multilayer switch
NetFlow

captures packets and saves them in a PCAP file
Wireshark

retrieves information on the operation of network devices
SNMP

Explanation: Topic 15.2.1

13. If a SOC has a goal of 99.999% uptime, how many minutes of downtime a year would be considered within its goal?

- Approximately 5 minutes per year.



- Approximately 10 minutes per year
- Approximately 20 minutes per year.
- Approximately 30 minutes per year.

Explanation: Topic 2.1.8

Within a year, there are 365 days x 24 hours a day x 60 minutes per hour = 525,600 minutes. With the goal of uptime 99.999% of time, the downtime needs to be controlled under $525,600 \times (1 - 0.99999) = 5.256$ minutes a year.

14. The HTTP server has responded to a client request with a 200 status code. What does this status code indicate?

- The request is understood by the server, but the resource will not be fulfilled.
- The request was completed successfully.
- The server could not find the requested resource, possibly because of an incorrect URL.
- The request has been accepted for processing, but processing is not completed.

Explanation: Topic 10.6.4

15. What is an advantage for small organizations of adopting IMAP instead of POP?



- POP only allows the client to store messages in a centralized way, while IMAP allows distributed storage.
- IMAP sends and retrieves email, but POP only retrieves email.
- When the user connects to a POP server, copies of the messages are kept in the mail server for a short time, but IMAP keeps them for a long time.
- Messages are kept in the mail servers until they are manually deleted from the email



client.

Explanation: Topic 10.5.4

IMAP and POP are protocols that are used to retrieve email messages. The advantage of using IMAP instead of POP is that when the user connects to an IMAP-capable server, copies of the messages are downloaded to the client application. IMAP then stores the email messages on the server until the user manually deletes those messages.

16. What debugging security tool can be used by black hats to reverse engineer binary files when writing exploits?

- WinDbg
- Firesheep
- Skipfish
- AIDE

Explanation: Topic 13.2.2

17. Match the attack tools with the description. (Not all options are used.)



Nmap	This is used for password cracking by either removing the original password, after bypassing the data encryption, or by outright discovery of the password. RainbowCrack This is a packet crafting tool used to probe and test the robustness of a firewall by using specially crafted, forged packets. Yersinia This is a wireless hacking tool used to detect security vulnerabilities in wireless networks. This is a network scanning tool used to probe network devices, servers, and hosts for open TCP or UDP ports. Nmap
Yersinia	
RainbowCrack	

Explanation: Topic 13.2.2

18. What are two features of ARP? (Choose two.)

- When a host is encapsulating a packet into a frame, it refers to the MAC address table to determine the mapping of IP addresses to MAC addresses.
- If a host is ready to send a packet to a local destination device and it has the IP address but not the MAC address of the destination, it generates an ARP broadcast.
- If a device receiving an ARP request has the destination IPv4 address, it responds with an ARP reply.
- If no device responds to the ARP request, then the originating node will broadcast the data packet to all devices on the network segment.
- An ARP request is sent to all devices on the Ethernet LAN and contains the IP address of the destination



Explanation: Topic 8.2

When a node encapsulates a data packet into a frame, it needs the destination MAC address. First it determines if the destination device is on the local network or on a remote network. Then it checks the ARP table (not the MAC table) to see if a pair of IP address and MAC address exists for either the destination IP address (if the destination host is on the local network) or the default gateway IP address (if the destination host is on a remote network). If the match does not exist, it generates an ARP broadcast to seek the IP address to MAC address resolution. Because the destination MAC address is unknown, the ARP request is broadcast with the MAC address FFFF.FFFF.FFFF. Either the destination device or the default gateway will respond with its MAC address, which enables the sending node to assemble the frame. If no device responds to the ARP request, then the originating node will discard the packet because a frame cannot be created.

19. What is a property of the ARP table on a device?

- Entries in an ARP table are time-stamped and are purged after the timeout expires.
- Every operating system uses the same timer to remove old entries from the ARP cache.
- Static IP-to-MAC address entries are removed dynamically from the ARP table.



for 3 minutes.

Explanation: Topic 8.2.6

20. What is the purpose of Tor?

- to allow users to browse the Internet anonymously
- to securely connect to a remote network over an unsecure link such as an Internet connection
- to donate processor cycles to distributed computational tasks in a processor sharing P2P network
- to inspect incoming traffic and look for any that violates a rule or matches the signature of a known exploit

Explanation: Topic 24.2.4

Tor is a software platform and network of peer-to-peer (P2P) hosts that function as routers. Users access the Tor network by using a special browser that allows them to browse anonymously.

21. Which two network protocols can be used by a threat actor to exfiltrate data in traffic that is disguised as normal network traffic? (Choose two.)



- NTP
- DNS
- HTTP
- syslog
- SMTP

Explanation: Topic 24.1.3



by NetFlow and data captured by Wireshark?

- NetFlow data shows network flow contents whereas Wireshark data shows network flow statistics.
- NetFlow data is analyzed by tcpdump whereas Wireshark data is analyzed by nfdump.
- NetFlow provides transaction data whereas Wireshark provides session data.
- NetFlow collects metadata from a network flow whereas Wireshark captures full data packets.

Explanation: Topic 25.1.3

Wireshark captures the entire contents of a packet. NetFlow does not. Instead, NetFlow collects metadata, or data about the flow.

23. Which tool captures full data packets with a command-line interface only?

- nfdump
- Wireshark
- NBAR2
- tcpdump

Explanation: Topic 25.3.1



Wireshark is a packet analyzer with a GUI interface.

24. Which method can be used to harden a device?

- maintain use of the same passwords
- allow default services to remain enabled
- allow USB auto-detection
- use SSH and disable the root account access over SSH

Explanation: Topic 4.4.2

The basic best practices for device hardening are as follows:

Ensure physical security.

Minimize installed packages.

Disable unused services.

Use SSH and disable the root account login over SSH.

Keep the system updated.

Disable USB auto-detection.

Enforce strong passwords.

Force periodic password changes.

Keep users from re-using old passwords.

Review logs regularly.

25. In a Linux operating system, which component interprets user commands and attempts to execute them?

- GUI
- daemon
- kernel
- shell

Explanation: Topic 4.2

26. A network administrator is configuring an AAA



(Choose two.)

- encryption for all communication
- encryption for only the data
- single process for authentication and authorization
- separate processes for authentication and authorization
- hidden passwords during transmission

Explanation: Topic 19.2.2

RADIUS authentication supports the following features:

RADIUS authentication and authorization as one process

Encrypts only the password

Utilizes UDP

Supports remote-access technologies, 802.1X, and Session Initiation Protocol (SIP)

27. What is privilege escalation?

- Vulnerabilities in systems are exploited to grant higher levels of privilege than someone or some process should have.
- Everyone is given full rights by default to everything and rights are taken away only when someone abuses privileges.



received a promotion.

- A security problem occurs when high ranking corporate officials demand rights to systems or files that they should not have.

Explanation: Topic 19.1.3

With privilege escalation, vulnerabilities are exploited to grant higher levels of privilege. After the privilege is granted, the threat actor can access sensitive information or take control of the system.

28. What two assurances does digital signing provide about code that is downloaded from the Internet? (Choose two.)

- The code contains no viruses.
- The code has not been modified since it left the software publisher.
- The code is authentic and is actually sourced by the publisher.
- The code contains no errors.
- The code was encrypted with both a private and public key.

Explanation: Topic 21.3.2

Digitally signing code provides several assurances about the code:

The code is authentic and is actually sourced by the publisher.

The code has not been modified since it left the software publisher.

The publisher undeniably published the code. This provides nonrepudiation of the act of publishing.



29. An IT enterprise is recommending the use of PKI applications to securely exchange information



exchange information between users? (Choose two.)

- HTTPS web service
- 802.1x authentication
- local NTP server
- FTP transfers
- file and directory access permission

Explanation: Topic 21.5.1

The Public Key Infrastructure (PKI) is a third party-system referred to as a certificate authority or CA. The PKI is the framework used to securely exchange information between parties. Common PKI applications are as follows:

SSL/TLS certificate-based peer authentication

Secure network traffic using IPsec VPNs

HTTPS Web traffic

Control access to the network using 802.1x authentication

Secure email using the S/MIME protocol

Secure instant messaging

Approve and authorize applications with Code Signing

Protect user data with the Encryption File System (EFS)

Implement two-factor authentication with smart cards

Securing USB storage devices

30. Which measure can a security analyst take to perform effective security monitoring against network traffic encrypted by SSL technology?

- Use a Syslog server to capture network traffic.
- Deploy a Cisco SSL Appliance.
- Require remote access connections through IPsec VPN.
- Deploy a Cisco ASA.

Explanation: Topic 21.5.3

Deploy a Cisco SSL Appliance to decrypt SSL traffic and send it to intrusion prevention system (IPS) appliances to



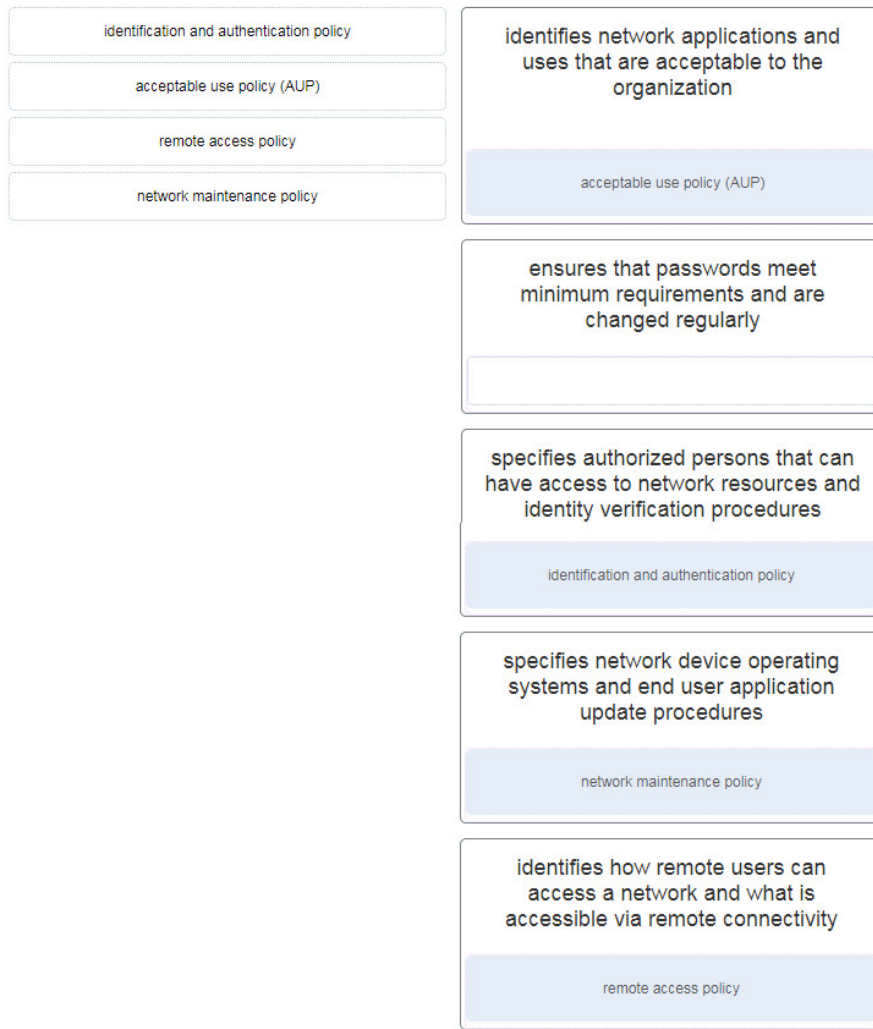
31. An administrator is trying to develop a BYOD security policy for employees that are bringing a wide range of devices to connect to the company network. Which three objectives must the BYOD security policy address? (Choose three.)

- All devices must be insured against liability if used to compromise the corporate network.
- All devices must have open authentication with the corporate network.
- Rights and activities permitted on the corporate network must be defined.
- Safeguards must be put in place for any personal device being compromised.
- The level of access of employees when connecting to the corporate network must be defined.
- All devices should be allowed to attach to the corporate network flawlessly.

Explanation: Topic 18.2.3

32. Match the security policy with the description. (Not all options are used.)





Explanation: Topic 18.2.2

identifies network applications and uses that are acceptable to the organization	acceptable use policy (AUP)
identifies how remote users can access a network and what is accessible via remote connectivity	remote access policy
specifies authorized persons that can have access to network resources and identity verification procedures	identification and authentication policy
specifies network device operating systems and end user application update procedures	network maintenance policy

33. Match the attack to the definition. (Not all options are used.)

attacker sends falsified information to redirect users to malicious sites	domain generation
attacker uses open resolvers to increase the volume of attacks and mask the true source of the attack	
attacker sends multiple packets that consume server resources	resource utilization attack
	attacker sends multiple packets that consume server resources
	ARP cache poisoning
	attacker sends falsified information to redirect users to malicious sites
	amplification and reflection
	attacker uses open resolvers to increase the volume of attacks and mask the true source of the attack



Explanation: Topic 17.1.3

attacker uses open resolvers to increase the volume of attacks and mask the true source of the attack	amplification and reflection
attacker sends falsified information to redirect users to malicious sites	ARP cache poisoning
attacker sends multiple packets that consume server resources	resource utilization attack

34. What type of attack targets an SQL database using the input field of a user?

- XML injection
- buffer overflow
- Cross-site scripting
- SQL injection

Explanation: Topic 17.2.4

A criminal can insert a malicious SQL statement in an entry field on a website where the system does not filter the user input correctly.

35. What are two characteristics of Ethernet MAC

- MAC addresses use a flexible hierarchical structure.
- They are expressed as 12 hexadecimal digits.
- They are globally unique.
- They are routable on the Internet.
- MAC addresses must be unique for both Ethernet and serial interfaces on a device.

Explanation: Topic 6.1.3

An Ethernet MAC address is a 48-bit binary value expressed as 12 hexadecimal digits. MAC addresses must be globally unique by design. MAC addresses are in flat structure and thus they are not routable on the Internet. Serial interfaces do not use MAC addresses.

36. A user calls to report that a PC cannot access the internet. The network technician asks the user to issue the command `ping 127.0.0.1` in a command prompt window. The user reports that the result is four positive replies. What conclusion can be drawn based on this connectivity test?

- The IP address obtained from the DHCP server is correct.
- The PC can access the network. The problem exists beyond the local network.
- The PC can access the Internet. However, the web browser may not work.
- The TCP/IP implementation is functional.

Explanation: Topic 7.2.3

The `ping 127.0.0.1` command is used to verify that the TCP/IP stack is functional. It verifies the proper operation of the protocol stack from the network layer to physical layer, without sending a signal on the media. That is, this test does not go beyond the PC itself. For example, it does not detect whether a cable is connected to the PC or not.



37. What characterizes a threat actor?

- They are all highly-skilled individuals.
- They always use advanced tools to launch attacks.
- They always try to cause some harm to an individual or organization.
- They all belong to organized crime.

Explanation: Topic 1.2.1

38. A computer is presenting a user with a screen requesting payment before the user data is allowed to be accessed by the same user. What type of malware is this?

- a type of logic bomb
- a type of virus
- a type of worm
- a type of ransomware

Explanation: Topic 14.1.7

Ransomware commonly encrypts data on a computer and makes the data unavailable until the computer user pays a specific sum of money



- router solicitation
- neighbor advertisement
- neighbor solicitation
- router advertisement

Explanation: Topic 7.1.2

IPv6-enabled router interfaces send periodic router advertisement messages that provide addressing information to hosts that perform SLAAC.

40. Which tool included in the Security Onion is a series of software plugins that send different types of data to the Elasticsearch data stores?

- Curator
- Beats
- OSSEC
- ElastAlert

Explanation: Topic 27.1.1

41. Which two types of unreadable network traffic could be eliminated from data collected by NSM? (Choose two.)



- IPsec traffic
- routing updates traffic
- SSL traffic
- broadcast traffic

Explanation: Topic 27.1.2

To reduce the huge amount of data collected so that cybersecurity analysts can focus on critical threats, some less important or unusable data could be eliminated from the datasets. For example, encrypted data, such as IPsec and SSL traffic, could be eliminated because it is unreadable in a reasonable time frame.

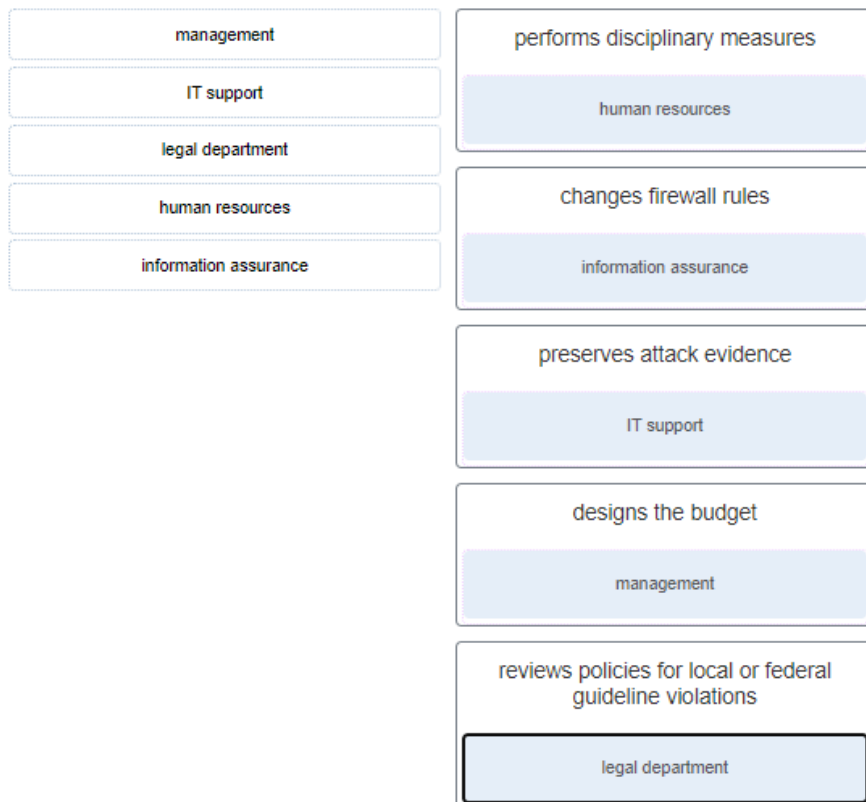
42. Which core open source component of the Elastic-stack is responsible for accepting the data in its native format and making elements of the data consistent across all sources?

- Logstash
- Kibana
- Beats
- Elasticsearch

Explanation: Topic 27.1.3

43. Match the security incident stakeholder with the role.





Explanation: Topic 28.4.3

44. In the NIST incident response process life cycle, which type of attack vector involves the use of brute force against devices, networks, or services?

- media
- impersonation
- attrition



Explanation: Topic 28.4.6

Common attack vectors include media, attrition, impersonation, and loss or theft. Attrition attacks are any attacks that use brute force. Media attacks are those initiated from storage devices. Impersonation attacks occur when something or someone is replaced for the purpose of the attack, and loss or theft attacks are initiated by equipment inside the organization.

45. Match the security organization with its security functions. (Not all options are used.)

Match the security organization with its security functions. (Not all options are used.)

SANS	It maintains and supports the Internet Storm Center and also develops security courses.
MITRE	
FIRST	
	SANS
	It maintains a list of common vulnerabilities and exposures (CVE).
	MITRE
	It provides vendor neutral educational products and career services to industry professionals globally.
	It brings together a variety of computer security incident response teams from government, commercial, and educational organizations to foster cooperation and coordination in information sharing, incident prevention and rapid reaction.
	FIRST

Explanation: Topic 20.1.1**46. What is a characteristic of CybOX?**

- It is a set of standardized schemata for specifying, capturing, characterizing, and communicating events and properties of



network operations.

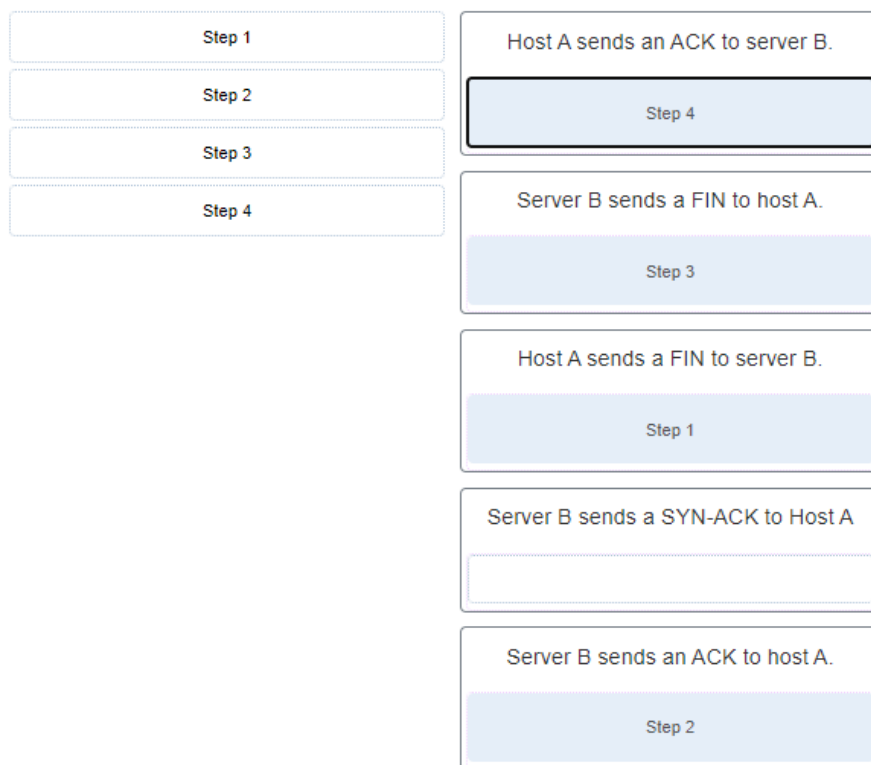
- It enables the real-time exchange of cyberthreat indicators between the U.S. Federal Government and the private sector.
- It is a set of specifications for exchanging cyberthreat information between organizations.
- It is the specification for an application layer protocol that allows the communication of CTI over HTTPS.

Explanation: Topic 20.2.5

CybOX is an open standards set of standardized schemata for specifying, capturing, characterizing, and communicating events and properties of network operations that support many cybersecurity functions.

47. After host A receives a web page from server B, host A terminates the connection with server B. Match each step to its correct option in the normal termination process for a TCP connection. (Not all options are used.)





Explanation: Topic 9.2.3

48. What are two ways that ICMP can be a security threat to a company? (Choose two.)

- by collecting information about a network
- by corrupting data between email servers and email recipients
- by the infiltration of web pages
- by corrupting network IP data packets
- by providing a conduit for DoS attacks

Explanation: Topic 16.2.2

ICMP can be used as a conduit for DoS attacks. It can be used to collect information about a network such as the identification of hosts and network structure, and by determining the operating systems being used on the network.

49. Which three IPv4 header fields have no equivalent



- fragment offset
- protocol
- flag
- TTL
- identification
- version

Explanation: Topic 16.1.4

Unlike IPv4, IPv6 routers do not perform fragmentation. Therefore, all three fields supporting fragmentation in the IPv4 header are removed and have no equivalent in the IPv6 header. These three fields are fragment offset, flag, and identification. IPv6 does support host packet fragmentation through the use of extension headers, which are not part of the IPv6 header.

50. Which two `net` commands are associated with network resource sharing? (Choose two.)



- net start
- net accounts
- net share
- net use
- net stop



The **net** command is a very important command. Some common **net** commands include these:

- **net accounts** – sets password and logon requirements for users
- **net session** – lists or disconnects sessions between a computer and other computers on the network
- **net share** – creates, removes, or manages shared resources
- **net start** – starts a network service or lists running network services
- **net stop** – stops a network service
- **net use** – connects, disconnects, and displays information about shared network resources
- **net view** – shows a list of computers and network devices on the network

51. Match the Windows 10 Registry key with its description. (Not all options are used.)

HKKEY_LOCAL_MACHINE	all of the configuration settings for the hardware and software configured on the computer for all users
HKKEY_CURRENT_USER	HKEY_USERS
HKKEY_CLASSES_ROOT	settings about the file system, file associations, shortcuts used when you ask Windows to run a file, or view a directory
HKEY_USERS	HKEY_CLASSES_ROOT
HKEY_CURRENT_CONFIG	data about the preferences of the currently logged on user, including personalization settings, default devices, and programs, etc
	HKEY_CURRENT_USER
	information about the current hardware profile of the machine
	HKEY_CURRENT_CONFIG



52. Which PDU format is used when bits are received from the network medium by the NIC of a host?

- segment
- file
- packet
- frame

Explanation: Topic 5.3.3

When received at the physical layer of a host, the bits are formatted into a frame at the data link layer. A packet is the PDU at the network layer. A segment is the PDU at the transport layer. A file is a data structure that may be used at the application layer.

53. A user is executing a tracert to a remote device. At what point would a router, which is in the path to the destination device, stop forwarding the packet?

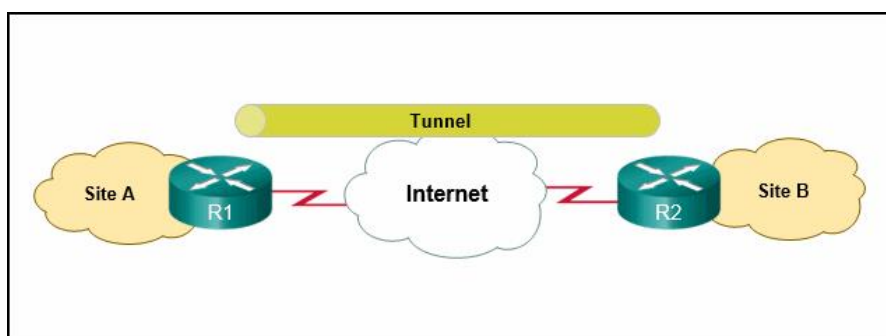
- when the router receives an ICMP Time Exceeded message
- when the values of both the Echo Request and Echo Reply messages reach zero
- when the RTT value reaches zero
- when the value in the TTL field reaches zero
- when the host responds with an ICMP Echo Reply



Explanation: Topic 7.2.6

When a router receives a traceroute packet, the value in the TTL field is decremented by 1. When the value in the field reaches zero, the receiving router will not forward the packet, and will send an ICMP Time Exceeded message back to the source.

54. Refer to the exhibit. What solution can provide a VPN between site A and site B to support encapsulation of any Layer 3 protocol between the internal networks at each site?



- an IPsec tunnel
- Cisco SSL VPN
- a GRE tunnel
- a remote access tunnel

Explanation: Topic 12.3.11

A Generic Routing Encapsulation (GRE) tunnel is a non-secure, site-to-site VPN tunneling solution that is capable of encapsulating any Layer 3 protocol between multiple sites across over an IP internetwork.

55. For what purpose would a network administrator use the Nmap tool?

- protection of the private IP addresses of internal hosts
- identification of specific network anomalies
- collection and analysis of security alerts and logs
- detection and identification of open ports



Explanation: Topic 13.2.1

Nmap allows an administrator to perform port scanning to probe computers and the network for open ports. This helps the administrator verify that network security policies are in place.

56. Match the network service with the description.

Categories:

Provides statistics on IP packets flowing through network devices

Notifies the administrator with detailed system messages

Allows administrators to manage network nodes

Synchronizes the time across all devices on the network

Options:

C SNMP

A NetFlow

D NTP

B syslog

A

B

C

D

A

B

C

D

Endpoint Security (ESec) Final Exam Answers 19

Explanation: Topic 12.4.1

provides statistics on IP packets flowing through network devices	NetFlow
notifies the administrator with detailed system messages	syslog
allows administrators to manage network nodes	SNMP
synchronizes the time across all devices on the network	NTP



58. Match the attack surface with attack exploits.

Match the attack surface with attack exploits.

Network Attack Surface	These attacks are delivered through exploitation of vulnerabilities in web, cloud, or host-based software applications. Software Attack Surface
Software Attack Surface	
Human Attack Surface	
	These attacks include conventional wired and wireless network protocols, as well as other wireless protocols used by smartphones or IoT devices. The attacks target vulnerabilities at the transport layer. Network Attack Surface
	These attacks include social engineering, malicious behaviour by trusted insiders, and user error. Human Attack Surface

Explanation: Topic 22.3.1

Software Attack Surface	These attacks are delivered through exploitation of vulnerabilities in web, cloud, or host-based software applications.
Network Attack Surface	These attacks include conventional wired and wireless network protocols, as well as other wireless protocols used by smartphones or IoT devices. The attacks target vulnerabilities at the transport layer.
Human Attack Surface	These attacks include social engineering, malicious behaviour by trusted insiders, and user error.

59. Match the Linux host-based firewall application



iptables

nftables

TCP Wrappers

This is a rule-based access control and logging system for Linux Packet filtering based on IP addresses and network services.

TCP Wrappers

This is an application that allows Linux system administrators to configure network access rules that are part of the Linux kernel Netfilter modules.

iptables

This application uses a simple virtual machine in the Linux kernel where code is executed and network packets are inspected.

nftables

Explanation: Topic 22.2.1

nftables	This application uses a simple virtual machine in the Linux kernel where code is executed and network packets are inspected.
iptables	This is an application that allows Linux system administrators to configure network access rules that are part of the Linux kernel Netfilter modules.
TCP Wrappers	This is a rule-based access control and logging system for Linux Packet filtering based on IP addresses and network services.



services.

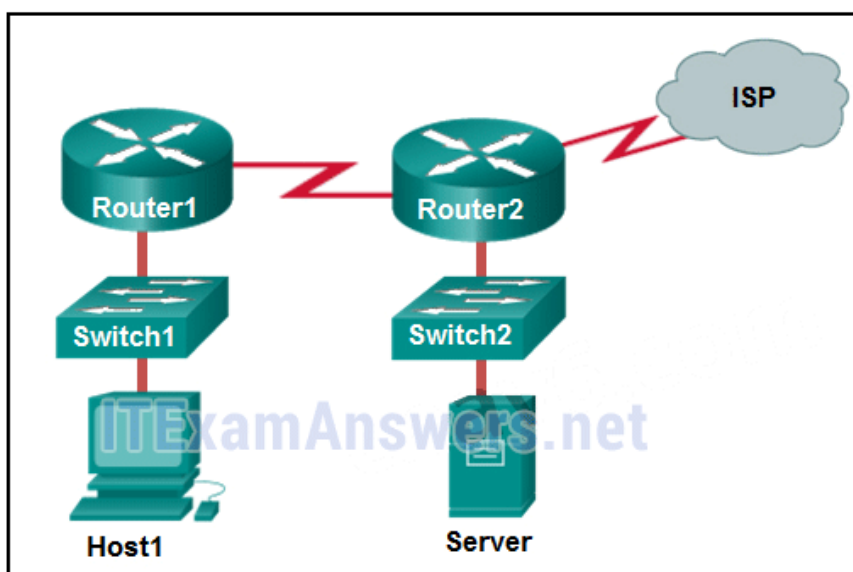
60. What network attack seeks to create a DoS for clients by preventing them from being able to obtain a DHCP lease?

- DHCP starvation
- IP address spoofing
- DHCP spoofing
- CAM table attack

Explanation: Topic 17.1.6

DCHP starvation attacks are launched by an attacker with the intent to create a DoS for DHCP clients. To accomplish this goal, the attacker uses a tool that sends many DHCPDISCOVER messages in order to lease the entire pool of available IP addresses, thus denying them to legitimate hosts.

61. Refer to the exhibit. If Host1 were to transfer a file to the server, what layers of the TCP/IP model would be used?



- only application and Internet layers
- application, transport, Internet, and network



access layers

- only Internet and network access layers
- only application, transport, network, data link, and physical layers
- only application, Internet, and network access layers
- application, session, transport, network, data link, and physical layers

Explanation: Topic 5.2.10

The TCP/IP model contains the application, transport, internet, and network access layers. A file transfer uses the FTP application layer protocol. The data would move from the application layer through all of the layers of the model and across the network to the file server.

62. A company has a file server that shares a folder named Public. The network security policy specifies that the Public folder is assigned Read-Only rights to anyone who can log into the server while the Edit rights are assigned only to the network admin group. Which component is addressed in the AAA network service framework?

- automation
- authentication
- authorization
- accounting

Explanation: Topic 12.3.10

After a user is successfully authenticated (logged into the server), the authorization is the process of determining what network resources the user can access and what operations (such as read or edit) the user can perform.

63. Match the destination network routing table entry type with a definition.



directly connected interface	found only in routers running IOS 15+ or IPv6 routing
dynamic route	local route interface
local route interface	automatically added when an interface is configured and active
static route	directly connected interface
	added when a protocol such as OSPF or EIGRP discovers a route
	dynamic route
	manually configured by a network administrator
	static route

Explanation: Topic 11.1.6

64. A person coming to a cafe for the first time wants to gain wireless access to the Internet using a laptop. What is the first step the wireless client will do in order to communicate over the network using a wireless management frame?

- associate with the AP
- authenticate to the AP
- discover the AP
- agree with the AP on the payload



Explanation: Topic 11.2.5

In order for wireless devices to communicate on a wireless network, management frames are used to complete a three-stage process:

Discover the AP

Authenticate with the AP

Associate with the AP

65. A device has been assigned the IPv6 address of 2001:0db8:cafe:4500:1000:00d8:0058:00ab/64. Which is the network identifier of the device?

- 2001:0db8:cafe:4500:1000
- 2001:0db8:cafe:4500:1000:00d8:0058:00ab
- 1000:00d8:0058:00ab
- 2001:0db8:cafe:4500
- 2001

Explanation: Topic 6.6.5

The address has a prefix length of /64. Thus the first 64 bits represent the network portion, whereas the last 64 bits represent the host portion of the IPv6 address.

66. An administrator wants to create four subnetworks from the network address 192.168.1.0/24. What is the network address and subnet mask of the second useable subnet?

subnetwork 192.168.1.64

subnet mask 255.255.255.192

subnetwork 192.168.1.64

subnet mask 255.255.255.240

subnetwork 192.168.1.32

subnet mask 255.255.255.240

subnetwork 192.168.1.128

subnet mask 255.255.255.192



Explanation: Topic 6.3.6

The number of bits that are borrowed would be two, thus giving a total of 4 useable subnets:

192.168.1.0

192.168.1.64

192.168.1.128

192.168.1.192

Because 2 bits are borrowed, the new subnet mask would be /26 or 255.255.255.192

67. What term describes a set of software tools designed to increase the privileges of a user or to grant access to the user to portions of the operating system that should not normally be allowed?

- compiler
- rootkit
- package manager
- penetration testing

Explanation: Topic 4.7.5

A rootkit is used by an attacker to secure a backdoor to a compromised computer, grant access to portions of the operating system normally not permitted, or increase the privileges of a user.



that the web server deployed in the DMZ is frequently targeted by threat actors. The decision is made to implement a patch management system to manage the server. Which risk management strategy method is being used to respond to the identified risk?

- risk sharing
- risk avoidance
- risk reduction
- risk retention

Explanation: Topic 23.3.1

There are four potential strategies for responding to risks that have been identified:

Risk avoidance – Stop performing the activities that create risk.

Risk reduction – Decrease the risk by taking measures to reduce vulnerability.

Risk sharing – Shift some of the risk to other parties.

Risk retention – Accept the risk and its consequences.

69. What are three characteristics of an information security management system? (Choose three.)

- It involves the implementation of systems that track the location and configuration of networked devices and software across an enterprise.
- It is a systematic and multilayered approach to cybersecurity.
- It addresses the inventory and control of hardware and software configurations of systems.
- It consists of a set of practices that are systematically applied to ensure continuous improvement in information security.



- It consists of a management framework through which an organization identifies, analyzes, and addresses information security risks.
- It is based on the application of servers and security devices.

Explanation: Topic 23.4

An Information Security Management System (ISMS) consists of a management framework through which an organization identifies, analyzes, and addresses information security risks. ISMSs are not based in servers or security devices. Instead, an ISMS consists of a set of practices that are systematically applied by an organization to ensure continuous improvement in information security. ISMSs provide conceptual models that guide organizations in planning, implementing, governing, and evaluating information security programs.

ISMSs are a natural extension of the use of popular business models, such as Total Quality Management (TQM) and Control Objectives for Information and Related Technologies (COBIT), into the realm of cybersecurity.

An ISMS is a systematic, multi-layered approach to cybersecurity. The approach includes people, processes, technologies, and the cultures in which they interact in a process of risk management.

70. Which three technologies should be included in a SOC security information and event management system? (Choose three.)

- event collection, correlation, and analysis
- security monitoring
- user authentication
- proxy service
- intrusion prevention
- threat intelligence



Explanation: Topic 2.1.4

Technologies in a SOC should include the following:

- Event collection, correlation, and analysis
- Security monitoring
- Security control
- Log management
- Vulnerability assessment
- Vulnerability tracking
- Threat intelligence

Proxy server, VPN, and IPS are security devices deployed in the network infrastructure.

71. What part of the URL, `http://www.cisco.com/index.html`, represents the top-level DNS domain?

- `http`
- `www`
- `.com`
- `index`



Explanation: Topic 10.2.2

The components of the URL `http://www.cisco.com/index.htm` are as follows:

`http` = protocol

`www` = part of the server name

`cisco` = part of the domain name

`index` = file name

`com` = the top-level domain

72. What best describes the security threat of spoofing?

- sending bulk email to individuals, lists, or domains with the intention to prevent users from accessing email
- sending abnormally large amounts of data to a remote server to prevent user access to the server services
- intercepting traffic between two hosts or inserting false information into traffic between two hosts
- making data appear to come from a source that is not the actual source

Explanation: Topic 14.2.4**73. A newly created company has fifteen Windows 10 computers that need to be installed before the company can open for business. What is a best practice that the technician should implement when configuring the Windows Firewall?**

- The technician should remove all default firewall rules and selectively deny traffic from reaching the company network.
- After implementing third party security software for the company, the technician should verify that the Windows Firewall is disabled.



users on how to allow an app through the Windows Firewall using the Administrator account.

- The technician should enable the Windows Firewall for inbound traffic and install other firewall software for outbound traffic control.

Explanation: Topic 3.4.6

Only disable Windows Firewall if other firewall software is installed. Use the Windows Firewall (Windows 7 or 8) or the Windows Defender Firewall (Windows 10) Control Panel to enable or disable the Windows Firewall.

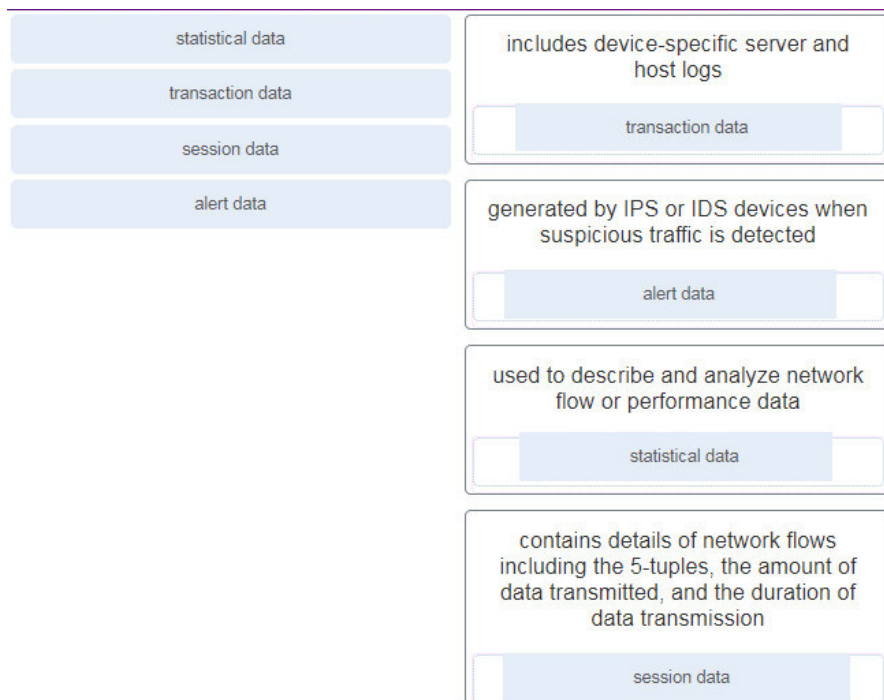
74. Which statement defines the difference between session data and transaction data in logs?

- Session data analyzes network traffic and predicts network behavior, whereas transaction data records network sessions.
- Session data is used to make predictions on network behaviors, whereas transaction data is used to detect network anomalies.
- Session data records a conversation between hosts, whereas transaction data focuses on the result of network sessions.
- Session data shows the result of a network session, whereas transaction data is in response to network threat traffic.

Explanation: Topic 25.1.2

75. Match the network monitoring data type with the description.





Explanation: Topic 25.1.1

76. Which device supports the use of SPAN to enable monitoring of malicious activity?

- Cisco Catalyst switch
- Cisco IronPort
- Cisco NAC
- Cisco Security Agent

Explanation: Topic 25.1.1



one port to be redirected to another port.

77. Which term is used for describing automated queries that are useful for adding efficiency to the cyberoperations workflow?

- cyber kill chain
- playbook
- chain of custody
- rootkit

Explanation: Topic 27.3.2

A playbook is an automated query that can add efficiency to the cyberoperations workflow.

78. When ACLs are configured to block IP address spoofing and DoS flood attacks, which ICMP message should be allowed both inbound and outbound?

- echo reply
- unreachable
- source quench
- echo

Explanation: Topic 24.2.1

Source quench ICMP messages provide the ability to have the sender throttle down the rate of messages when necessary. These messages should be allowed through the firewall in both inbound and outbound directions.

79. After a security monitoring tool identifies a malware attachment entering the network, what is the benefit of performing a retrospective analysis?



- It can identify how the malware originally entered the network.
- A retrospective analysis can help in tracking the behavior of the malware from the identification point forward.
- It can calculate the probability of a future incident.
- It can determine which network host was first affected.

Explanation: Topic 24.1.5

General security monitoring can identify when a malware attachment enters a network and which host is first infected. Retrospective analysis takes the next step and is the tracking of the behavior of the malware from that point forward.

80. Which two data types would be classified as personally identifiable information (PII)? (Choose two.)

- house thermostat reading
- average number of cattle per region
- vehicle identification number
- hospital emergency use per region
- Facebook photographs

Explanation: Topic 1.3.1

81. A help desk technician notices an increased number of calls relating to the performance of computers located at the manufacturing plant. The



What are two purposes of botnets? (Choose two.)

- to transmit viruses or spam to computers on the same network
- to record any and all keystrokes
- to attack other computers
- to withhold access to a computer or files until money has been paid
- to gain access to the restricted part of the operating system

Explanation: Topic 14.3.2

Botnets can be used to perform DDoS attacks, obtain data, or transmit malware to other devices on the network.

82. Which two statements describe the use of asymmetric algorithms? (Choose two.)

- Public and private keys may be used interchangeably.
- If a public key is used to encrypt the data, a private key must be used to decrypt the data.
- If a public key is used to encrypt the data, a public key must be used to decrypt the data.
- If a private key is used to encrypt the data, a public key must be used to decrypt the data.
- If a private key is used to encrypt the data, a private key must be used to decrypt the data.

Explanation: Topic 21.2.4

Asymmetric algorithms use two keys: a public key and a private key. Both keys are capable of the encryption process, but the complementary matched key is required for decryption. If a public key encrypts the data, the matching private key decrypts the data. The opposite is also true. If a private key encrypts the data, the corresponding public key decrypts the data.



83. Which three security services are provided by digital signatures? (Choose three.)

- provides nonrepudiation using HMAC functions
- guarantees data has not changed in transit
- provides data encryption
- authenticates the source
- provides confidentiality of digitally signed data
- authenticates the destination

Explanation: Topic 21.3.1

Digital signatures are a mathematical technique used to provide three basic security services. Digital signatures have specific properties that enable entity authentication and data integrity. In addition, digital signatures provide nonrepudiation of the transaction. In other words, the digital signature serves as legal proof that the data exchange did take place.

84. What are two methods to maintain certificate revocation status? (Choose two.)

- CRL
- DNS
- subordinate CA
- OCSP



- LDAP

Explanation: Topic 21.4.6

A digital certificate might need to be revoked if its key is compromised or it is no longer needed. The certificate revocation list (CRL) and Online Certificate Status Protocol (OCSP), are two common methods to check a certificate revocation status.

**85. What are two uses of an access control list?
(Choose two.)**

- ACLs provide a basic level of security for network access.
- ACLs can control which areas a host can access on a network.
- Standard ACLs can restrict access to specific applications and ports.
- ACLs assist the router in determining the best path to a destination.
- ACLs can permit or deny traffic based upon the MAC address originating on the router.

Explanation: Topic 12.3.2

ACLs can be used for the following:

- Limit network traffic in order to provide adequate network performance
- Restrict the delivery of routing updates
- Provide a basic level of security
- Filter traffic based on the type of traffic being sent
- Filter traffic based on IP addressing

86. A client is using SLAAC to obtain an IPv6 address for the interface. After an address has been generated and applied to the interface, what must the client do before it can begin to use this IPv6 address?

- It must send an ICMPv6 Router Solicitation message



- It must send an ICMPv6 Router Solicitation message to request the address of the DNS server.
- It must send an ICMPv6 Neighbor Solicitation message to ensure that the address is not already in use on the network.
- It must wait for an ICMPv6 Router Advertisement message giving permission to use this address.

Explanation: Topic 7.1.2

Stateless DHCPv6 or stateful DHCPv6 uses a DHCP server, but Stateless Address Autoconfiguration (SLAAC) does not. A SLAAC client can automatically generate an address that is based on information from local routers via Router Advertisement (RA) messages. Once an address has been assigned to an interface via SLAAC, the client must ensure via Duplicate Address Detection (DAD) that the address is not already in use. It does this by sending out an ICMPv6 Neighbor Solicitation message and listening for a response. If a response is received, then it means that another device is already using this address.

87. A technician is troubleshooting a network connectivity problem. Pings to the local wireless router are successful but pings to a server on the Internet are unsuccessful. Which CLI command could assist the technician to find the location of the networking problem?



- tracert
- ipconfig
- msconfig
- ipconfig/renew

Explanation: Topic 7.2.6

The tracert utility (also known as the tracert command or tracert tool) will enable the technician to locate the link to the server that is down. The ipconfig command displays the computer network configuration details. The ipconfig/renew command requests an IP address from a DHCP server. Msconfig is not a network troubleshooting command.

88. What are two evasion techniques that are used by hackers? (Choose two.)

- Trojan horse
- pivot
- rootkit
- reconnaissance
- phishing

Explanation: Topic 14.3.6

The following methods are used by hackers to avoid detection: Encryption and tunneling – hide or scramble the malware content

Resource exhaustion – keeps the host device too busy to detect the invasion

Traffic fragmentation – splits the malware into multiple packets

Protocol-level misinterpretation – sneaks by the firewall

Pivot – uses a compromised network device to attempt access to another device

Rootkit – allows the hacker to be undetected and hides software installed by the hacker



mitigate a compromised system during the Actions on Objectives step as defined by the Cyber Kill Chain model? (Choose two.)

- Perform forensic analysis of endpoints for rapid triage.
- Train web developers for securing code.
- Build detections for the behavior of known malware.
- Collect malware files and metadata for future analysis.
- Detect data exfiltration, lateral movement, and unauthorized credential usage.

Explanation: Topic 28.2.8

When security professionals are alerted about the system compromises, forensic analysis of endpoints should be performed immediately for rapid triage. In addition, detection efforts for further attacking activities such as data exfiltration, lateral movement, and unauthorized credential usage should be enhanced to reduce damage to the minimum.

90. Place the seven steps defined in the Cyber Kill Chain in the correct order.





Explanation: Topic 28.2.1

91. Which field in the TCP header indicates the status of the three-way handshake process?

- control bits



- reserved
- checksum

Explanation: Topic 9.2.4

The value in the control bits field of the TCP header indicates the progress and status of the connection.

92. A user opens three browsers on the same PC to access www.cisco.com to search for certification course information. The Cisco web server sends a datagram as a reply to the request from one of the web browsers. Which information is used by the TCP/IP protocol stack in the PC to identify which of the three web browsers should receive the reply?

- the source IP address
- the destination port number
- the destination IP address
- the source port number

Explanation: Topic 9.1.2

Each web browser client application opens a randomly generated port number in the range of the registered ports and uses this number as the source port number in the datagram that it sends to a server. The server then uses this port number as the destination port number in the reply datagram that it sends to the web browser. The PC that is running the web browser application receives the datagram and uses the destination port number that is contained in this datagram to identify the client application.



93. What are two scenarios where probabilistic security analysis is best suited? (Choose two.)

- when applications that conform to application/networking standards are analyzed
- when random variables create difficulty in knowing with certainty the outcome of any



given event

- when each event is the inevitable result of antecedent causes
- when analyzing applications designed to circumvent firewalls
- when analyzing events with the assumption that they follow predefined steps

Explanation: Topic 26.2.3

Probabilistic analysis relies on statistical techniques that are designed to estimate the probability that an event will occur based on the likelihood that prior events will occur.

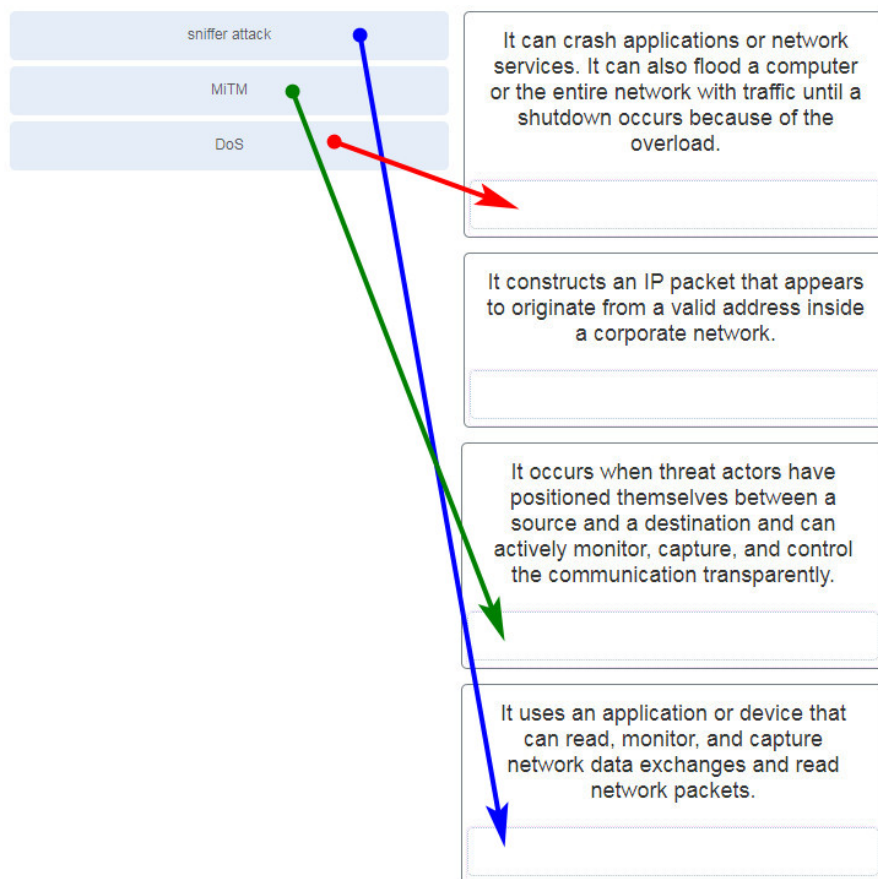
94. Which tool is a web application that provides the cybersecurity analyst an easy-to-read means of viewing an entire Layer 4 session?

- Snort
- Zeek
- CapME
- OSSEC

Explanation: Topic 26.1.2

95. Match the category of attacks with the description. (Not all options are used.)





Explanation: Topic 13.2.3

MiTM	It occurs when threat actors have positioned themselves between a source and a destination and can actively monitor, capture, and control the communication transparently.
------	--



sniffer attack	It uses an application or device that can read, monitor, and capture network data exchanges and read network packets.
DoS	It can crash applications or network services. It can also flood a computer or the entire network with traffic until a shutdown occurs because of the overload.

96. What are two characteristics of the SLAAC method for IPv6 address configuration? (Choose two.)

- The default gateway of an IPv6 client on a LAN will be the link-local address of the router interface attached to the LAN.
- This stateful method of acquiring an IPv6 address requires at least one DHCPv6 server.
- Clients send router advertisement messages to routers to request IPv6 addressing.
- IPv6 addressing is dynamically assigned to clients through the use of ICMPv6.
- Router solicitation messages are sent by the router to offer IPv6 addressing to clients.

Explanation: Topic 7.1.2

With SLAAC, the default gateway for IPv6 clients will be the link-local address of the router interface that is attached to the client LAN. The IPv6 addressing is dynamically assigned via the ICMPv6 protocol. SLAAC is a stateless method of acquiring an IPv6 address, a method that requires no servers. When a client is configured to obtain its addressing information automatically via SLAAC, the client sends a router solicitation message to the IPv6 all-routers multicast address FF02::2. The router advertisement messages are sent by routers to provide addressing information to clients.



97. A technician notices that an application is not responding to commands and that the computer seems to respond slowly when applications are opened. What is the best administrative tool to force the release of system resources from the unresponsive application?

- Event Viewer
- System Restore
- Add or Remove Programs
- Task Manager

Explanation: Topic 3.3.6

Use the Task Manager Performance tab to see a visual representation of CPU and RAM utilization. This is helpful in determining if more memory is needed. Use the Applications tab to halt an application that is not responding.

98. How can statistical data be used to describe or predict network behavior?

- by comparing normal network behavior to current network behavior
- by recording conversations between network endpoints
- by listing results of user web surfing activities
- by displaying alert messages that are generated by Snort

Explanation: Topic 25.1.4

Statistical data is created through the analysis of other forms of network data. Statistical characteristics of normal network behavior can be compared to current network traffic in an effort to detect anomalies. Conclusions resulting from analysis can be used to describe or predict network behavior.



used with an attack vector?

- the proximity of the threat actor to the vulnerability
- the presence or absence of the requirement for user interaction in order for an exploit to be successful
- the determination whether the initial authority changes to a second authority during the exploit
- the number of components, software, hardware, or networks, that are beyond the control of the attacker and that must be present in order for a vulnerability to be successfully exploited

Explanation: Topic 23.2.3

This is a metric that reflects the proximity of the threat actor to the vulnerable component. The more remote the threat actor is to the component, the higher the severity. Threat actors close to your network or inside your network are easier to detect and mitigate.

**100. Which NIST Cybersecurity Framework core function is concerned with the development and implementation of safeguards that ensure the delivery of critical infrastructure services?**

- respond
- detect



- recover
- protect

Explanation: Topic 23.4.3

101. Which two techniques are used in a smurf attack? (Choose two.)

- session hijacking
- resource exhaustion
- botnets
- amplification
- reflection

Explanation: Topic 16.2.4

A smurf attack uses amplification and reflection techniques to overwhelm a targeted host. The threat actor forwards ICMP echo request messages that contain the source IP address of the victim to a large number of hosts. These hosts all reply to the spoofed IP address of the victim with the intent of overwhelming it.

102. What is the primary objective of a threat intelligence platform (TIP)?

- to aggregate the data in one place and present it in a comprehensible and usable format
- to provide a specification for an application layer protocol that allows the communication of CTI over HTTPS
- to provide a standardized schema for specifying, capturing, characterizing, and communicating events and properties of network operations
- to provide a security operations platform that



Explanation: Topic 20.2.6

A threat intelligence platform (TIP) centralizes the collection of threat data from numerous data sources and formats. TIP is designed to aggregate the data in one place and present it in a comprehensible and usable format. This is especially important as the volume of threat intelligence data can be overwhelming.

103. Which wireless parameter is used by an access point to broadcast frames that include the SSID?

- security mode
- active mode
- passive mode
- channel setting

Explanation: Topic 11.2.6

The two scanning or probing modes an access point can be placed into are passive or active. In passive mode, the AP advertises the SSID, supported standards, and security settings in broadcast beacon frames. In active mode, the wireless client must be manually configured for the same wireless parameters as the AP has configured.

104. Match the field in the Event table of Sguil to the description.

Match the field in the Event table of Sguil to the description

Explanation: Topic 11.2.6

105. An employee connects wirelessly to the company network using a cell phone. The employee then configures the cell phone to act as a wireless access



describes this situation?

- rogue access point
- cracking
- denial of service
- spoofing

Explanation: Topic 1.1.1

Configuring the cell phone to act as a wireless access point means that the cell phone is now a rogue access point. The employee unknowingly breached the security of the company network by allowing a user to access the network without connecting through the company access point. Cracking is the process of obtaining passwords from data stored or transmitted on a network. Denial of service attacks refer to sending large amounts of data to a networked device, such as a server, to prevent legitimate access to the server. Spoofing refers to access gained to a network or data by an attacker appearing to be a legitimate network device or user.

106. What information is required for a WHOIS query?

- outside global address of the client
- ICANN lookup server address
- link-local address of the domain owner
- FQDN of the domain

Explanation: Topic 10.2.6**107. Which two statements describe the characteristics of symmetric algorithms? (Choose two.)**

- They are referred to as a pre-shared key or secret key.
- They use a pair of a public key and a private key.
- They are commonly used with VPN traffic.
- They provide confidentiality, integrity, and availability.

Explanation: Topic 21.2.2

Symmetric encryption algorithms use the same key (also called shared secret) to encrypt and decrypt the data. In contrast, asymmetric encryption algorithms use a pair of keys, one for encryption and another for decryption.

108. What are two drawbacks to using HIPS? (Choose two.)

- With HIPS, the success or failure of an attack cannot be readily determined.
- With HIPS, the network administrator must verify support for all the different operating systems used in the network.
- HIPS has difficulty constructing an accurate network picture or coordinating events that occur across the entire network.
- If the network traffic stream is encrypted, HIPS is unable to access unencrypted forms of the traffic.
- HIPS installations are vulnerable to fragmentation attacks or variable TTL attacks

Explanation: Topic 12.2.7

The following are two drawbacks of HIPS:



events that might be occurring beyond an individual host and (2) every host operating system within the organization must be supported. However, an advantage of using HIPS is that it can monitor and protect the operating system as well as critical system processes on each network host.

109. What are three functions provided by the syslog service? (Choose three.)

- to select the type of logging information that is captured
- to periodically poll agents for data
- to provide statistics on packets that are flowing through a Cisco device
- to provide traffic analysis
- to gather logging information for monitoring and troubleshooting
- to specify the destinations of captured messages

Explanation: Topic 12.3.8

There are three primary functions provided by the syslog service:

1. gathering logging information
2. selection of the type of information to be logged
3. selection of the destination of the logged information



110. Which consideration is important when implementing syslog in a network?

- Enable the highest level of syslog available to ensure logging of all possible event messages.
- Synchronize clocks on all network devices with a protocol such as Network Time Protocol.
- Log all messages to the system buffer so that they



- Use SSH to access syslog information

Explanation: Topic 12.3.8

**111. What are the two ways threat actors use NTP?
(Choose two.)**

- They place an attachment inside an email message.
- They attack the NTP infrastructure in order to corrupt the information used to log the attack.
- They place iFrames on a frequently used corporate web page.
- They encode stolen data as the subdomain portion where the nameserver is under control of an attacker.
- Threat actors use NTP systems to direct DDoS attacks.

Explanation: Topic 24.1.2

Threat actors may attempt to attack the NTP infrastructure in order to corrupt time information used to correlate logged network events. This can serve to obfuscate traces of ongoing exploits. In addition, threat actors have been known to use NTP systems to direct DDoS attacks through vulnerabilities in client or server software. While these attacks do not necessarily result in corrupted security monitoring data, they can disrupt network availability.



112. Which two features are included by both TACACS+ and RADIUS protocols? (Choose two.)

- password encryption
- separate authentication and authorization processes
- SIP support
- utilization of transport layer protocols
- 802.1X support

Explanation: Topic 12.3.10

Both TACACS+ and RADIUS support password encryption (TACACS+ encrypts all communication) and use Layer 4 protocol (TACACS+ uses TCP and RADIUS uses UDP).

TACACS+ supports separation of authentication and authorization processes, while RADIUS combines authentication and authorization as one process. RADIUS supports remote access technology, such as 802.1x and SIP; TACACS+ does not.

113. Match the SIEM function to the description.

Match the SIEM function to the description.

forensic analysis	reduces the volume of event data by consolidating duplicate event records
correlation	aggregation
aggregation	presents event data in real-time monitoring and long-time summaries
reporting	reporting
	speeds detection of and reaction to security threats by examining logs and events from different systems
	correlation
	searches logs and events from sources throughout the organization for complete information analysis
	forensic analysis



Explanation: Topic 15.2.4

114. What are two types of attacks used on DNS open resolvers? (Choose two.)

- amplification and reflection
- fast flux
- ARP poisoning
- resource utilization
- cushioning

Explanation: Topic 17.1.3

Three types of attacks used on DNS open resolvers are as follows: DNS cache poisoning – attacker sends spoofed falsified information to redirect users from legitimate sites to malicious sites

DNS amplification and reflection attacks – attacker sends an increased volume of attacks to mask the true source of the attack

DNS resource utilization attacks – a denial of service (DoS) attack that consumes server resources

115. What are three goals of a port scan attack? (Choose three.)



- to identify peripheral configurations



- to determine potential vulnerabilities
- to disable used ports and services
- to identify operating systems
- to identify active services
- to discover system passwords

Explanation: Topic 14.2.2

116. Which protocol or service uses UDP for a client-to-server communication and TCP for server-to-server communication?

- HTTP
- FTP
- DNS
- SMTP

Explanation: Topic 10.2.3

Some applications may use both TCP and UDP. DNS uses UDP when clients send requests to a DNS server, and TCP when two DNS servers directly communicate.

117. What is one difference between the client-server and peer-to-peer network models?

- Only in the client-server model can file transfers occur.
- A data transfer that uses a device serving in a client role requires that a dedicated server be present.
- A peer-to-peer network transfers data faster than a transfer using a client-server network.
- Every device in a peer-to-peer network can function as a client or a server.



including the amount of traffic, the quality of service imposed, and the network media. Transfer speeds are not dependent on the network model type. File transfers can occur using the client-server model or the peer-to-peer model. A data transfer between a device acting in the client role and a device acting in the server role can occur in both peer-to-peer and client-server networks.

118. Which statement is correct about network protocols?

- They define how messages are exchanged between the source and the destination.
- They all function in the network access layer of TCP/IP.
- They are only required for exchange of messages between devices on remote networks.
- Network protocols define the type of hardware that is used and how it is mounted in racks.

Explanation: Topic 5.4.1

Network protocols are implemented in hardware, or software, or both. They interact with each other within different layers of a protocol stack. Protocols have nothing to do with the installation of the network equipment.

Network protocols are required to exchange information between source and destination devices in both local and remote networks.



119. Which approach can help block potential malware delivery methods, as described in the Cyber Kill Chain model, on an Internet-faced web server?



- Build detections for the behavior of known malware.
- Collect malware files and metadata for future analysis.
- Audit the web server to forensically determine the origin of exploit.
- Analyze the infrastructure storage path used for files.

Explanation: Topic 28.2.4

A threat actor may send the weapon through web interfaces to the target server, either in file uploads or coded web requests. By analyzing the infrastructure storage path used for files, security measures can be implemented to monitor and detect malware deliveries through these methods.

120. Which meta-feature element in the Diamond Model classifies the general type of intrusion event?



- phase
- results
- methodology
- direction

Explanation: Topic 28.3.1

Methodology - This is used to classify the general type of event, such as port scan, phishing, content delivery attack, syn flood, etc.



121. Which Linux command is used to manage processes?

- chrootkit
- ls
- grep
- kill

Explanation: Topic 4.7.3

The kill command is used to stop, restart, or pause a process. The chrootkit command is used to check the computer for rootkits, a set of software tools that can increase the privilege level of a user or grant access to portions of software normally not allowed. The grep command is used to look for a file or text within a file. The ls command is used to list files, directories, and file information.

122. Which tool can be used in a Cisco AVC system to analyze and present the application analysis data into dashboard reports?

- NetFlow
- NBAR2
- Prime
- IPFIX

Explanation: Topic 25.3.3

A management and reporting system, such as Cisco Prime, can be used to analyze and present the application analysis data into dashboard reports for use by network monitoring personnel.

123. Which Windows Event Viewer log includes events regarding the operation of drivers, processes, and hardware?



- system logs
- application logs
- security logs
- setup logs

Explanation: Topic 25.2.1

By default Windows keeps four types of host logs:

- **Application logs** – events logged by various applications
- **System logs** – events about the operation of drivers, processes, and hardware
- **Setup logs** – information about the installation of software, including Windows updates
- **Security logs** – events related to security, such as logon attempts and operations related to file or object management and access

124. Which method is used to make data unreadable to unauthorized users?

- Encrypt the data.
- Fragment the data.
- Add a checksum to the end of the data.
- Assign it a username and password.

Explanation: Topic 19.1.1

Network data can be encrypted using various cryptography



cryptography application so the data can be unencrypted.

125. Match the tabs of the Windows 10 Task Manager to their functions. (Not all options are used.)

Categories:

- Displays resource utilization information for CPU, memory, network, disk, and others
- Allows programs that are running on system startup to be disabled
- Allows for a start, stop or restart of a particular service
- Allows for a process to have its affinity set

Options:

- Services
- Startup
- Details
- Performance

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Diagram showing connections between categories and options:

- A (Performance) connects to C (Services)
- B (Startup) connects to B (Startup)
- C (Details) connects to D (Details)
- D (Affinity) connects to A (Performance)

Explanation: Topic 3.3.6

Details	Allows for a process to have its affinity set.
Performance	Displays resource utilization information for CPU, memory, network, disk, and others.
Startup	Allows programs that are running on system startup to be disabled.
Services	Allows for a start, stop or restart of a particular service.

126. For network systems, which management system addresses the inventory and control of hardware and software configurations?

- asset management
- vulnerability management
- risk management
- configuration management

Explanation: Topic 23.3.6

Configuration management addresses the inventory and



systems.

127. Match the common network technology or protocol with the description. (Not all options are used.)

NTP	uses application protocols that are commonly responsible for bringing malware to a host
Syslog	
ICMP	
DNS	

uses a hierarchy of authoritative time sources to send time information between devices on the network	NTP
used by attackers to exfiltrate data in traffic disguised as normal client queries	DNS
uses UDP port 514 for logging event messages from network devices and endpoints	Syslog
used by attackers to identify hosts on a network and the structure of the network	ICMP



Explanation: Topic 24.1.1

128. What are the three core functions provided by the Security Onion? (Choose three.)

- business continuity planning
- full packet capture
- alert analysis
- intrusion detection
- security device management
- threat containment

Explanation: Topic 26.1.1

Security Onion is an open source suite of Network Security Monitoring (NSM) tools for evaluating cybersecurity alerts. For cybersecurity analysts the Security Onion provides full packet capture, network-based and host-based intrusion detection systems, and alert analysis tools.

129. In NAT terms, what address type refers to the globally routable IPv4 address of a destination host on the Internet?

- outside global
- inside global
- outside local
- inside local

Explanation: Topic 10.3.4

From the perspective of a NAT device, inside global addresses are used by external users to reach internal hosts. Inside local addresses are the addresses assigned to internal hosts. Outside global addresses are the addresses of destinations on the external network. Outside local addresses are the actual private addresses of destination



130. Which two fields or features does Ethernet examine to determine if a received frame is passed to the data link layer or discarded by the NIC? (Choose two.)

- CEF
- source MAC address
- minimum frame size
- auto-MDIX
- Frame Check Sequence

Explanation: Topic 6.1.2

An Ethernet frame is not processed and is discarded if it is smaller than the minimum (64 bytes) or if the calculated frame check sequence (FCS) value does not match the received FCS value. Auto-MDIX (automatic medium-dependent interface crossover) is Layer 1 technology that detects cable straight-through or crossover types. The source MAC address is not used to determine how the frame is received. CEF (Cisco Express Forwarding) is a technology used to expedite Layer 3 switching.

131. Which type of data would be considered an example of volatile data?

- web browser cache
- memory registers



- log files
- temp files

Explanation: Topic 28.1.6

Volatile data is data stored in memory such as registers, cache, and RAM, or it is data that exists in transit. Volatile memory is lost when the computer loses power.

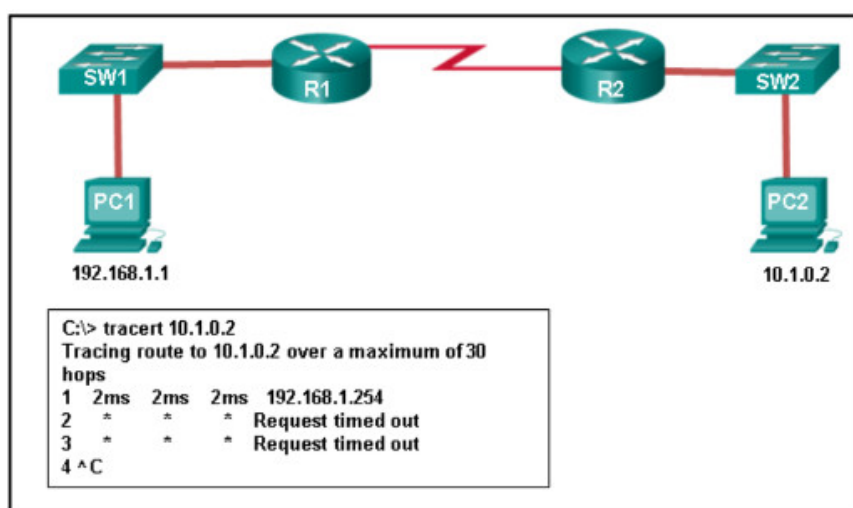
132. What is the main purpose of exploitations by a threat actor through the weapon delivered to a target during the Cyber Kill Chain exploitation phase?

- Launch a DoS attack.
- Send a message back to a CnC controlled by the threat actor.
- Break the vulnerability and gain control of the target.
- Establish a back door into the system.

Explanation: Topic 28.2.5

After the weapon has been delivered, the threat actor uses it to break the vulnerability and gain control of the target. The threat actor will use an exploit that gains the effect desired, does it quietly, and avoids detections. Establishing a back door in the target system is the phase of installation.

133. Refer to the exhibit. An administrator is trying to troubleshoot connectivity between PC1 and PC2 and uses the tracert command from PC1 to do it. Based on the displayed output, where should the administrator begin troubleshooting?



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- R1
- PC2
- SW2
- R2
- SW1

Explanation: Topic 7.2.6

Tracert is used to trace the path a packet takes. The only successful response was from the first device along the path on the same LAN as the sending host. The first device is the default gateway on router R1. The administrator should therefore start troubleshooting at R1.

134. What three security tools does Cisco Talos maintain security incident detection rule sets for?
(Choose three.)

- Snort
- NetStumbler
- Socat
- SpamCop
- ClamAV



the Snort.org, ClamAV, and SpamCop network security tools.

135. Which host-based firewall uses a three-profile approach to configure the firewall functionality?

- Windows Firewall
- iptables
- TCP Wrapper
- nftables

Explanation: Topic 22.2.1

Windows Firewall uses a profile-based approach to configuring firewall functionality. It uses three profiles, Public, Private, and Domain, to define firewall functions.

136. When a user visits an online store website that uses HTTPS, the user browser queries the CA for a CRL. What is the purpose of this query?

- to verify the validity of the digital certificate
- to request the CA self-signed digital certificate
- to check the length of key used for the digital certificate
- to negotiate the best encryption to use



A digital certificate must be revoked if it is invalid. CAs maintain a certificate revocation list (CRL), a list of revoked certificate serial numbers that have been invalidated. The user browser will query the CRL to verify the validity of a certificate.

137. Which step in the Vulnerability Management Life Cycle determines a baseline risk profile to eliminate risks based on asset criticality, vulnerability threat, and asset classification?

- discover
- assess
- prioritize assets
- verify

Explanation: Topic 23.3.3

The steps in the Vulnerability Management Life Cycle include these:

- Discover – inventory all assets across the network and identify host details, including operating systems and open services, to identify vulnerabilities
- Prioritize assets – categorize assets into groups or business units, and assign a business value to asset groups based on their criticality to business operations
- Assess – determine a baseline risk profile to eliminate risks based on asset criticality, vulnerability threats, and asset classification
- Report – measure the level of business risk associated with assets according to security policies. Document a security plan, monitor suspicious activity, and describe known vulnerabilities.
- Remediate – prioritize according to business risk and fix vulnerabilities in order of risk
- Verify – verify that threats have been eliminated through follow-up audits



138. Which management system implements systems



devices and software across an enterprise?

- asset management
- vulnerability management
- risk management
- configuration management

Explanation: Topic 23.3.4

Asset management involves the implementation of systems that track the location and configuration of networked devices and software across an enterprise.

139. A network administrator is reviewing server alerts because of reports of network slowness. The administrator confirms that an alert was an actual security incident. What is the security alert classification of this type of scenario?

- false negative
- true positive
- true negative
- false positive

Explanation: Topic 26.2.2

True Positive: The alert has been verified to be an actual security incident.



incident. Benign activity that results in a false positive is sometimes referred to as a benign trigger.

True Negative: No security incident has occurred. The activity is benign.

False Negative: An undetected incident has occurred.

140. Which application layer protocol is used to provide file-sharing and print services to Microsoft applications?

- SMTP
- HTTP
- SMB
- DHCP

Explanation: Topic 10.4.2

SMB is used in Microsoft networking for file-sharing and print services. The Linux operating system provides a method of sharing resources with Microsoft networks by using a version of SMB called SAMBA.

141. Which device in a layered defense-in-depth approach denies connections initiated from untrusted networks to internal networks, but allows internal users within an organization to connect to untrusted networks?

- access layer switch
- firewall
- internal router
- IPS

Explanation: Topic 18.1.4

A firewall is typically a second line of defense in a layered defense-in-depth approach to network security. The firewall typically connects to an edge router that connects to the service provider. The firewall tracks connections initiated



initiation of connections from external untrusted networks going to internal trusted networks.

142. What are two potential network problems that can result from ARP operation? (Choose two.)

- Large numbers of ARP request broadcasts could cause the host MAC address table to overflow and prevent the host from communicating on the network.
- On large networks with low bandwidth, multiple ARP broadcasts could cause data communication delays.
- Network attackers could manipulate MAC address and IP address mappings in ARP messages with the intent of intercepting network traffic.
- Multiple ARP replies result in the switch MAC address table containing entries that match the MAC addresses of hosts that are connected to the relevant switch port.
- Manually configuring static ARP associations could facilitate ARP poisoning or MAC address spoofing.

Explanation: Topic 8.3.1

Large numbers of ARP broadcast messages could cause momentary data communications delays. Network attackers could manipulate MAC address and IP address mappings in ARP messages with the intent to intercept network traffic. ARP requests and replies cause entries to be made into the ARP table, not the MAC address table. ARP table overflows are very unlikely. Manually configuring static ARP associations is a way to prevent, not facilitate, ARP poisoning and MAC address spoofing. Multiple ARP replies resulting in the switch MAC address table containing entries that match the MAC addresses of connected nodes and are associated with the relevant switch port are required for normal switch frame forwarding operations. It is not an ARP



143. Which three procedures in Sguil are provided to security analysts to address alerts? (Choose three.)

- Escalate an uncertain alert.
- Correlate similar alerts into a single line.
- Categorize true positives.
- Pivot to other information sources and tools.
- Construct queries using Query Builder.
- Expire false positives.

Explanation: Topic 27.2.4

Sguil is a tool for addressing alerts. Three tasks can be completed in Sguil to manage alerts:

- Alerts that have been found to be false positives can be expired.
- An alert can be escalated if the cybersecurity analyst is uncertain how to handle it.
- Events that have been identified as true positives can be categorized.



144. Match the SOC metric with the description. (Not all options apply.)



MTTD	The average time that it takes for the SOC personnel to identify that valid security incidents have occurred in the network.
MTTC	
MTTR	
	The time required to stop the incident from causing further damage to systems or data.
	The average time that it takes to stop and remediate a security incident.
	The average length of time that threat actors have access to a network before they are detected and their access is stopped.

Explanation: Topic 2.1.6

145. Which two services are provided by the NetFlow tool? (Choose two.)

- QoS configuration
- usage-based network billing
- log analysis
- access list monitoring
- network monitoring

Explanation: Topic 12.3.6

NetFlow efficiently provides an important set of services for IP applications including network traffic accounting, usage-based network billing, network planning, security, denial of service monitoring capabilities, and network monitoring.

146. An administrator discovers that a user is accessing a newly established website that may be



the administrator take first in terms of the security policy?

- Ask the user to stop immediately and inform the user that this constitutes grounds for dismissal.
- Create a firewall rule blocking the respective website.
- Revise the AUP immediately and get all users to sign the updated AUP.
- Immediately suspend the network privileges of the user.

Explanation: Topic 18.2.2

One of the most common security policy components is an AUP. This can also be referred to as an appropriate use policy. This component defines what users are allowed and not allowed to do on the various system components. This includes the type of traffic that is allowed on the network.

The AUP should be as explicit as possible to avoid misunderstanding.

147. Which two tasks can be performed by a local DNS server? (Choose two.)

- allowing data transfer between two network devices
- retrieving email messages
- providing IP addresses to local hosts
- forwarding name resolution requests between servers



- mapping name-to-IP addresses for internal hosts

Explanation: Topic 10.2.1

Two important functions of DNS are to (1) provide IP addresses for domain names such as <http://www.cisco.com>, and (2) forward requests that cannot be resolved to other servers in order to provide domain name to IP address translation. DHCP provides IP addressing information to local devices. A file transfer protocol such as FTP, SFTP, or TFTP provides file sharing services. IMAP or POP can be used to retrieve an email message from a server.

148. Which type of event is logged in Cisco Next-Generation IPS devices (NGIPS) using FirePOWER Services when changes have been detected in the monitored network?

- intrusion
- connection
- host or endpoint
- network discovery

Explanation: Topic 25.3.7

Network discovery events in Cisco NGIPS represent changes that have been detected in the monitored network.



149. Which two actions should be taken during the preparation phase of the incident response life cycle defined by NIST? (Choose two.)

- Acquire and deploy the tools that are needed to investigate incidents.
- Detect all the incidents that occurred.
- Meet with all involved parties to discuss the incident that took place.
- Create and train the CSIRT.



- Fully analyze the incident.

Explanation: Topic 28.4.4

According to the guideline defined in the NIST Incident Response Life Cycle, several actions should be taken during the preparation phase including (1) creating and training the CSIRT and (2) acquiring and deploying the tools needed by the team to investigate incidents.

150. What subnet mask is represented by the slash notation /20?

- 255.255.255.0
- 255.255.255.248
- 255.255.255.192
- 255.255.240.0
- 255.255.224.0

Explanation: Topic 6.3.3

The slash notation /20 represents a subnet mask with 20 1s. This would translate to: 11111111.11111111.11110000.0000, which in turn would convert into 255.255.240.0.

151. What is the benefit of converting log file data into a common schema?

source

- creates a set of regex-based field extractions
- allows the implementation of partial normalization and inspection
- allows easy processing and analysis of datasets

Explanation: Topic 27.1.3

When data is converted into a universal format, it can be effectively structured for performing fast queries and event analysis.

152. Which Cisco sponsored certification is designed to provide the first step in acquiring the knowledge and skills to work with a SOC team?

- CCNA CyberOps Associate
- CCNA Cloud
- CCNA Security
- CCNA Data Center

Explanation: Topic 2.2.1

The Cisco Certified CyberOps Associate certification provides a valuable first step in acquiring the knowledge and skills needed to work with a SOC team. It can be a valuable part of a career in the exciting and growing field of cybersecurity operations.



153. Which three IP addresses are considered private addresses? (Choose three.)

- 198.168.6.18
- 192.168.5.29
- 172.68.83.35
- 128.37.255.6
- 172.17.254.4



- 10.234.2.1

Explanation: Topic 10.3.1

The designated private IP addresses are within the three IP address ranges:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

154. When establishing a network profile for an organization, which element describes the time between the establishment of a data flow and its termination?

- bandwidth of the Internet connection
- routing protocol convergence
- session duration
- total throughput

Explanation: Topic 23.1.1

A network profile should include some important elements, such as the following:

Total throughput – the amount of data passing from a given source to a given destination in a given period of time

Session duration – the time between the establishment of a



available to accept data

Critical asset address space – the IP addresses or the logical location of essential systems or data

155. What are the stages that a wireless device completes before it can communicate over a wireless LAN network?

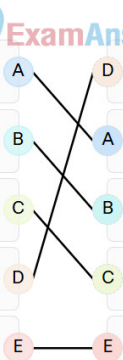
- discover a wireless AP, authenticate with the AP, associate with the AP
- discover a wireless AP, associate with the AP, authorize with the AP
- discover a wireless AP, associate with the AP, authenticate with the AP
- discover a wireless AP, authorize with the AP, associate with the AP

Explanation: Topic 11.2.5

156. Match the correct sequence of steps typically taken by a threat actor carrying out a domain shadowing attack.

Categories:  **Options:**

Step 1	A	D	An exploit kit landing page is created
Step 2	B	A	The website is compromised
Step 3	C	B	HTTP 302 cushioning is used
Step 4	D	C	Domain shadowing is used
Step 5	E	E	Malware is spread through its payload



Explanation: Topic 17.2.2

Step 1: The website is compromised

Step 2: HTTP 302 cushioning is used

Step 3: Domain shadowing is used

Step 4: An exploit kit landing page is created

Step 5: Malware is spread through its payload



157. What are two properties of a cryptographic hash function? (Choose two.)

- Complex inputs will produce complex hashes.
- The output is a fixed length.
- The hash function is one way and irreversible.
- Hash functions can be duplicated for authentication purposes.
- The input for a particular hash algorithm has to have a fixed size.

Explanation: Topic 21.1.2

A cryptographic hash function should have the following properties: The input can be any length.

The output has a fixed length.

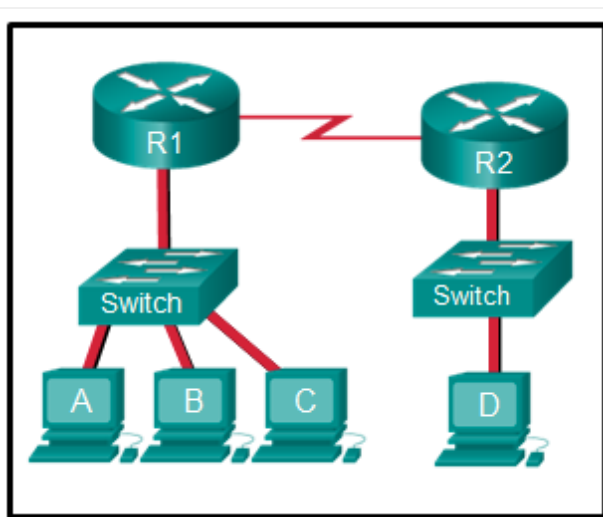
The hash value is relatively easy to compute for any given input.

The hash is one way and not reversible.

The hash is collision free, meaning that two different input values will result in different hash values

158. Refer to the exhibit. The switches have a default configuration. Host A needs to communicate with host D, but host A does not have the MAC address for the default gateway. Which network devices will receive the ARP request sent by host A?





i360201v3n1_275353.png

- only host D
- only hosts A, B, C, and D
- only hosts B and C
- only hosts B, C, and router R1
- only hosts A, B, and C
- only router R1

Explanation: Topic 8.3.1

Because host A does not have the MAC address of the default gateway in the ARP table, host A sends an ARP broadcast. The ARP broadcast would be sent to every device on the local network. Hosts B, C, and router R1 would receive the broadcast. Router R1 would not forward the message.

**159. Which type of evidence cannot prove an IT security fact on its own?**

- hearsay
- corroborative
- best
- indirect

Explanation: Topic 28.1.4

information. Best evidence is most reliable because it is something concrete such as a signed contract.

160. What is a characteristic of a probabilistic analysis in an alert evaluation?

- each event an inevitable result of antecedent causes
- precise methods that yield the same result every time by relying on predefined conditions
- random variables that create difficulty in knowing the outcome of any given event with certainty
- analysis of applications that conform to application/networking standards

Explanation: Topic 26.2.3

Statistical techniques can be used to evaluate the risk that exploits will be successful in a given network. This type of analysis can help decision makers to better evaluate the cost of mitigating a threat and the damage that an exploit could cause. Two general approaches used to do this are as follows:

- **Deterministic Analysis:** For an exploit to be successful, all prior steps in the exploit must also be successful. The cybersecurity analyst knows the steps for a successful exploit.
- **Probabilistic Analysis:** Statistical techniques are used to determine the probability that a successful exploit will occur based on the likelihood that each step in the exploit will succeed.



161. Why would a network administrator choose Linux as an operating system in the Security Operations Center (SOC)?

- It is easier to use than other server operating systems.
- It can be acquired at no charge.



- The administrator has control over specific security functions, but not standard applications.

Explanation: Topic 4.1.2

Linux is open source - Any person can acquire Linux at no charge and modify it to fit specific needs. This flexibility allows analysts and administrators to tailor-build an operating system specifically for security analysis.

162. A technician needs to verify file permissions on a specific Linux file. Which command would the technician use?

- cd
- sudo
- ls -l
- vi

Explanation: Topic 4.5.2

163. Which two protocols may devices use in the application process that sends email? (Choose two.)

- HTTP
- POP
- POP3
- DNS



- IMAP
- SMTP

Explanation: Topic 10.5

POP, POP3, and IMAP are protocols that are used to retrieve email from servers. SMTP is the default protocol that is used to send email. DNS may be used by the sender email server to find the address of the destination email server. HTTP is a protocol for send and receiving web pages.

164. Which file system type was specifically created for optical disk media?

- ext3
- HFS+
- CDFS
- ext2

Explanation: Topic 4.5.1**165. A piece of malware has gained access to a workstation and issued a DNS lookup query to a CnC server. What is the purpose of this attack?**

- to check the domain name of the workstation
- to send stolen sensitive data with encoding
- to masquerade the IP address of the workstation
- to request a change of the IP address

Explanation: Topic 24.1.3

A piece of malware, after accessing a host, may exploit the DNS service by communicating with command-and-control (CnC) servers and then exfiltrate data in traffic disguised as normal DNS lookup queries. Various types of encoding, such as base64, 8-bit binary, and hex can be used to



(DLP) measures.

166. Refer to the exhibit. Which field in the Sguil event window indicates the number of times an event is detected for the same source and destination IP address?

ST	CNT	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
RT	8	seconion-ossec	1.13	2017-06-19 23:10:40	0.0.0.0		0.0.0.0	0		[OSSEC] Integrity checks
RT	232	seconion-ossec	1.24	2017-06-19 23:18:28	0.0.0.0		0.0.0.0	0		[OSSEC] Received 0 pack
RT	6	seconion-ossec	1.3	2017-06-19 23:08:51	0.0.0.0		0.0.0.0	0		[OSSEC] Interface enter
RT	1	seconion-ossec	1.41	2017-06-30 14:34:56	0.0.0.0		0.0.0.0	0		[OSSEC] User login failed
RT	3	seconion-ossec	1.42	2017-06-30 14:39:31	0.0.0.0		0.0.0.0	0		[OSSEC] Integrity checks
RT	3	seconion-ossec	1.52	2017-06-30 15:04:27	0.0.0.0		0.0.0.0	0		[OSSEC] Host-based an
RT	1	seconion-ossec	1.7	2017-06-19 23:09:11	0.0.0.0		0.0.0.0	0		[OSSEC] New group add
RT	16	seconion-ossec	1.8	2017-06-19 23:09:26	0.0.0.0		0.0.0.0	0		[OSSEC] Integrity checks
RT	6	seconion-eth1-1	5.1	2017-06-19 23:18:22	209.165.200.235		192.168.0.1	1		GPL ICMP_INFO PING B
RT	10	seconion-eth1-1	5.13	2017-06-19 23:38:49	209.165.200.226		209.165.200.235	1		GPL ICMP_INFO PING *
RT	6	seconion-eth1-1	5.2	2017-06-19 23:18:22	209.165.200.235		192.168.0.1	1		GPL ICMP_INFO PING *
RT	1	seconion-eth1-1	5.23	2017-06-19 23:51:12	209.165.201.17	40599	209.165.200.235	21		ET EXPLOIT VSFTPD Back
RT	1	seconion-eth1-1	5.24	2017-06-19 23:51:12	209.165.200.235	6200	209.165.201.17	34057	6	GPL ATTACK_RESPONSE
RT	106	seconion-eth2-1	7.1	2017-06-19 23:19:00	209.165.201.17		192.168.0.1	1		GPL ICMP_INFO PING *
RT	51	seconion-eth2-1	7.6	2017-06-19 23:39:03	209.165.201.21		209.165.201.17	1		GPL ICMP_INFO PING *
RT	1	seconion-eth2-1	7.90	2017-06-19 23:51:12	209.165.201.17	40599	209.165.200.235	21	6	ET EXPLOIT VSFTPD Back
RT	1	seconion-eth2-1	7.91	2017-06-19 23:51:12	209.165.200.235	6200	209.165.201.17	34057	6	GPL ATTACK_RESPONSE

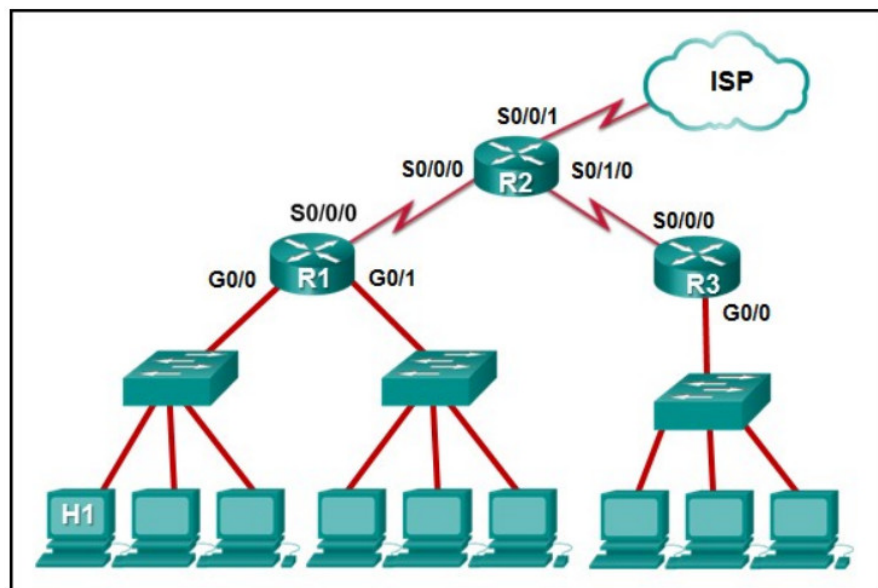
- CNT
- Pr
- ST
- AlertID

Explanation: Topic 26.1.4

The CNT field indicates the number of times an event is detected from the same source and destination IP address. Having a high number of events can indicate a problem with event signatures.



device interface should be used as the default gateway setting of host H1?



- R1: G0/0
- R2: S0/0/0
- R2: S0/0/1
- R1: S0/0/0

Explanation: Topic 6.5.2

The default gateway for host H1 is the router interface that is attached to the LAN that H1 is a member of. In this case, that is the G0/0 interface of R1. H1 should be configured with the IP address of that interface in its addressing settings. R1 will provide routing services to packets from H1 that need to be forwarded to remote networks.

168. According to information outlined by the Cyber Kill Chain, which two approaches can help identify reconnaissance threats? (Choose two.)

- Analyze web log alerts and historical search data.
- Audit endpoints to forensically determine origin of exploit.
- Build playbooks for detecting browser behavior.



- Conduct full malware analysis.
- Understand targeted servers, people, and data available to attack.

Explanation: Topic 28.2.2

Threat actors may use port scanning toward a web server of an organization and identify vulnerabilities on the server. They may visit the web server to collect information about the organization. The web server logging should be enabled and the logging data should be analyzed to identify possible reconnaissance threats. Building playbooks by filtering and combining related web activities by visitors can sometimes reveal the intentions of threat actors.

169. Which two ICMPv6 messages are used during the Ethernet MAC address resolution process? (Choose two.)

- router solicitation
- router advertisement
- neighbor solicitation
- neighbor advertisement
- echo request

Explanation: Topic 11.1.5

IPv6 uses neighbor solicitation (NS) and neighbor advertisement (NA) ICMPv6 messages for MAC address resolution.

170. What best describes the destination IPv4 address that is used by multicasting?



- a single IP multicast address that is used by all destinations in a group
- an IP address that is unique for each destination in the group
- a group address that shares the last 23 bits with the source IPv4 address
- a 48 bit address that is determined by the number of members in the multicast group

Explanation: Topic 5.2.7

The destination multicast IPv4 address is a group address, which is a single IP multicast address within the Class D range.

171. What is the result of using security devices that include HTTPS decryption and inspection services?



- The devices require continuous monitoring and fine tuning.
- The devices introduce processing delays and privacy issues.
- The devices must have preconfigured usernames and passwords for all users.
- Monthly service contracts with reputable web filtering sites can be costly.



HTTPS adds extra overhead to the HTTP-formed packet. HTTPS encrypts using Secure Sockets Layer (SSL). Even though some devices can perform SSL decryption and inspection, this can present processing and privacy issues.

172. What is a disadvantage of DDNS?

- DDNS is considered malignant and must be monitored by security software.
- DDNS is unable to co-exist on a network subdomain that also uses DNS.
- Using free DDNS services, threat actors can quickly and easily generate subdomains and change DNS records.
- Using DDNS, a change in an existing IP address mapping can take over 24 hours and could result in a disruption of connectivity.

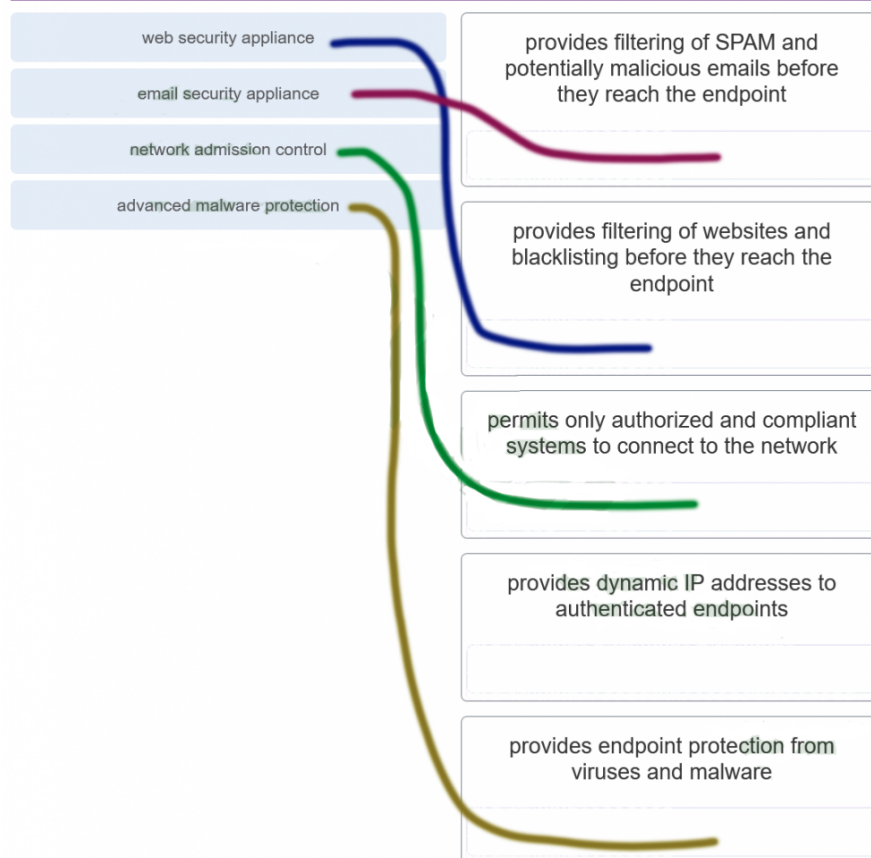
Explanation: Topic 10.2.5

Dynamic DNS can be abused by threat actors in various ways. Free DDNS services are especially useful to threat actors. DDNS can be used to facilitate the rapid change of IP address for malware command-and-control servers after the current IP address has become widely blocked.

173. Match the network-based antimalware solution to the function. (Not all options are used.)



Match the network-based antimalware solution to the function. (Not all options are used.)



Explanation: Topic 22.1.4

174. A threat actor has identified the potential vulnerability of the web server of an organization and is building an attack. What will the threat actor possibly do to build an attack weapon?

- Obtain an automated tool in order to deliver



the malware payload through the vulnerability.

- Install a webshell on the web server for persistent access.
- Create a point of persistence by adding services.
- Collect credentials of the web server developers and administrators.

Explanation: Topic 28.2.3

One tactic of weaponization used by a threat actor after the vulnerability is identified is to obtain an automated tool to deliver the malware payload through the vulnerability.

175. Which tool included in the Security Onion is a series of software plugins that send different types of data to the Elasticsearch data stores?

- OSSEC
- Curator
- Beats
- ElastAlert

Explanation: Topic 27.1.1

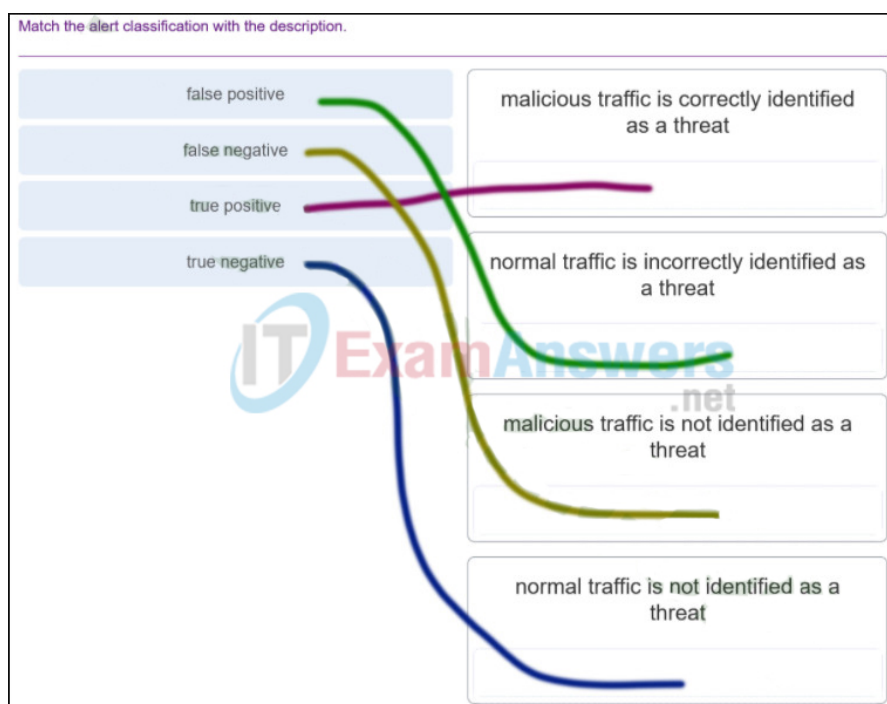
Beats – This is a series of software plugins that send different types of data to the Elasticsearch data stores.

176. Which term is used to describe the process of identifying the NSM-related data to be gathered?

- data archiving
- data normalization
- data reduction
- data retention

Explanation: Topic 27.1.2





Explanation: Topic 26.2.2

178. According to NIST, which step in the digital forensics process involves preparing and presenting information that resulted from scrutinizing data?

- examination
- collection
- reporting
- analysis



NIST describes the digital forensics process as involving the following four steps:

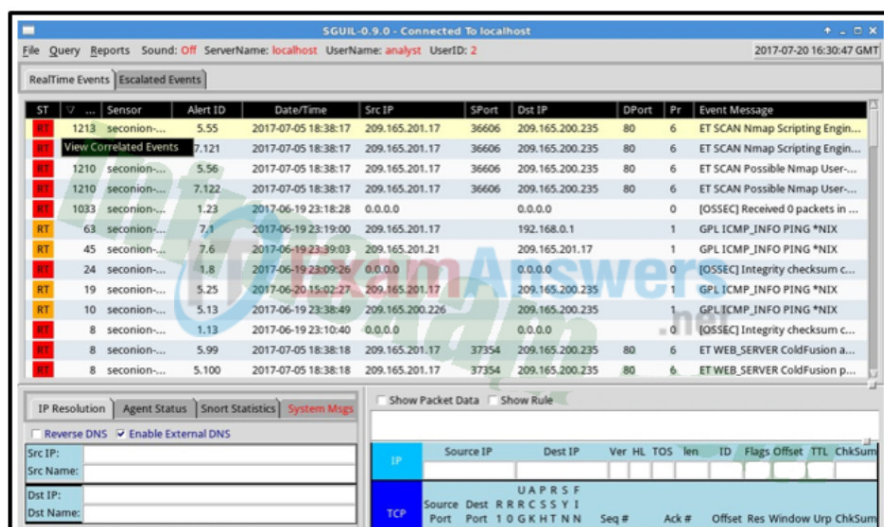
Collection – the identification of potential sources of forensic data and acquisition, handling, and storage of that data

Examination – assessing and extracting relevant information from the collected data. This may involve decompression or decryption of the data

Analysis – drawing conclusions from the data. Salient features, such as people, places, times, events, and so on should be documented

Reporting – preparing and presenting information that resulted from the analysis. Reporting should be impartial and alternative explanations should be offered if appropriate

179. Refer to the exhibit. A cybersecurity analyst is using Sguil to verify security alerts. How is the current view sorted?



ST	✓	Sensor	Alert ID	Date/Time	Src IP	SPort	Dst IP	DPort	Pr	Event Message
1213		seconion...	5.55	2017-07-05 18:38:17	209.165.201.17	36606	209.165.200.235	80	6	ET SCAN Nmap Scripting Engin...
View Correlated Events										
1210		seconion...	5.56	2017-07-05 18:38:17	209.165.201.17	36606	209.165.200.235	80	6	ET SCAN Possible Nmap User...
1210		seconion...	7.122	2017-07-05 18:38:17	209.165.201.17	36606	209.165.200.235	80	6	ET SCAN Possible Nmap User...
1033		seconion...	1.23	2017-06-19 23:18:28	0.0.0.0	0.0.0.0	0.0.0.0	0	0	[OSSEC] Received 0 packets in ...
63		seconion...	7.1	2017-06-19 23:19:00	209.165.201.17	192.168.0.1	1	1	1	GPL ICMP_INFO PING *NIX
45		seconion...	7.6	2017-06-19 23:39:03	209.165.201.21	209.165.201.17	1	1	1	GPL ICMP_INFO PING *NIX
24		seconion...	1.8	2017-06-19 23:09:26	0.0.0.0	0.0.0.0	0	0	0	[OSSEC] Integrity checksum c...
19		seconion...	5.25	2017-06-20 15:02:27	209.165.201.17	209.165.200.235	1	1	1	GPL ICMP_INFO PING *NIX
10		seconion...	5.13	2017-06-19 23:38:49	209.165.200.226	209.165.200.235	1	1	1	GPL ICMP_INFO PING *NIX
8		seconion...	1.13	2017-06-19 23:10:40	0.0.0.0	0.0.0.0	0	0	0	[OSSEC] Integrity checksum c...
8		seconion...	5.99	2017-07-05 18:38:18	209.165.201.17	37354	209.165.200.235	80	6	ET WEB_SERVER ColdFusion a...
8		seconion...	5.100	2017-07-05 18:38:18	209.165.201.17	37354	209.165.200.235	80	6	ET WEB_SERVER ColdFusion p...

- by sensor number
- by source IP
- by date/time
- by frequency

Explanation: Topic 27.2.1

The CNT column, between the ST and Sensor columns, displays the frequency of alerts. By sorting with frequency, the analyst will get a better sense of what has happened on the network.

180. Which two options are window managers for Linux? (Choose two.)

- File Explorer
- Kali
- Gnome
- PenTesting
- KDE

Explanation: Topic 4.6

The X Window System provides the basic framework for a GUI, but the GUI itself varies greatly between different distributions. Two window managers are Gnome and KDE.

181. What are the two methods that a wireless NIC can use to discover an AP? (Choose two.)

- transmitting a probe request
- sending an ARP request broadcast
- initiating a three-way handshake
- receiving a broadcast beacon frame
- sending a multicast frame

Explanation: Topic 14.2.2



and register with an access point: passive mode and active mode. In passive mode, the AP sends a broadcast beacon frame that contains the SSID and other wireless settings. In active mode, the wireless device must be manually configured for the SSID, and then the device broadcasts a probe request.

182. A client device has initiated a secure HTTP request to a web browser. Which well-known port address number is associated with the destination address?

- 110
- 80
- 443
- 404

Explanation: Topic 10.6.6

Port numbers are used in TCP and UDP communications to differentiate between the various services running on a device. The well-known port number used by HTTPs is port 443.

183. Which term describes evidence that is in its original state?



- Corroborating evidence
- Best evidence



- Indirect evidence
- Direct evidence

Explanation: Topic 28.1.4

Evidence can be classified as follows:

Best evidence: This is evidence that is in its original state. It might be storage devices used by an accused or archives of files that can be proven to be unaltered.

Corroborating evidence: This is evidence that supports a proposition already supported by initial evidence, therefore confirming the original proposition.

Indirect evidence: This evidence acts in combination with other facts to establish a hypothesis.

184. Which three statements describe a DHCP Discover message? (Choose three.)

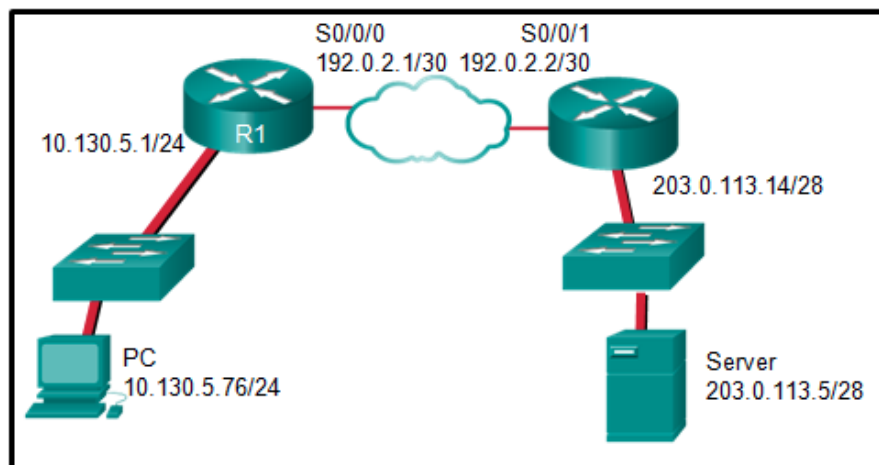
- The source MAC address is 48 ones (FF-FF-FF-FF-FF-FF).
- The destination IP address is 255.255.255.255.
- The message comes from a server offering an IP address.
- The message comes from a client seeking an IP address.
- All hosts receive the message, but only a DHCP server replies.
- Only the DHCP server receives the message.

Explanation: Topic 10.1.2

When a host configured to use DHCP powers up on a network it sends a DHCPDISCOVER message. FF-FF-FF-FF-FF-FF is the L2 broadcast address. A DHCP server replies with a unicast DHCPOFFER message back to the host.

185. Refer to the exhibit. The PC is sending a packet to

performing NAT overload. From the perspective of the PC, match the NAT address type with the correct IP address. (Not all options are used.)



Inside global	203.0.113.5
Outside global	
Inside local	203.0.113.14
	192.0.2.2
	192.0.2.1
	10.130.5.1
	10.130.5.76

Arrows indicate the following matches:

- Inside global points to 203.0.113.5
- Outside global points to 192.0.2.1
- Inside local points to 10.130.5.76

Place the options in the following order:



Inside global	192.0.2.1 
Inside local	10.130.5.76 

Explanation: Topic 24.2.2

The inside local address is the private IP address of the source or the PC in this instance. The inside global address is the translated address of the source or the address as seen by the outside device. Since the PC is using the outside address of the R1 router, the inside global address is 192.0.2.1. The outside addressing is simply the address of the server or 203.0.113.5.

186. Which step in the Vulnerability Management Life Cycle categorizes assets into groups or business units, and assigns a business value to asset groups based on their criticality to business operations?

- remediate
- prioritize assets
- report
- assess

Explanation: Topic 23.3.3

The steps in the Vulnerability Management Life Cycle include these:

Discover – inventory all assets across the network and identify host details, including operating systems and open services to identify vulnerabilities

Prioritize assets – categorize assets into groups or business units, and assign a business value to asset groups based on their criticality to business operations

Assess – determine a baseline risk profile to eliminate risks based on asset criticality, vulnerability threats, and asset classification

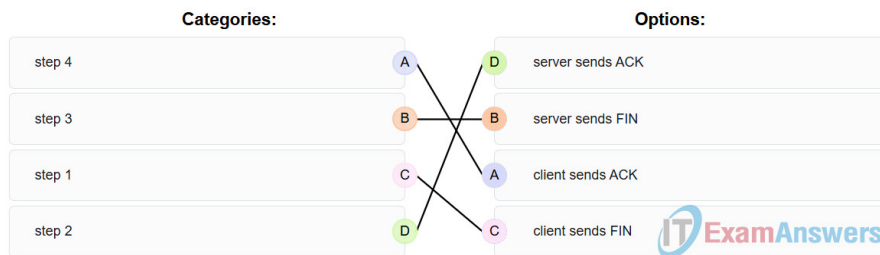
Report – measure the level of business risk associated with your assets according to your security policies. Document a security plan, monitor suspicious activity, and describe



Remediate – prioritize according to business risk and fix vulnerabilities in order of risk

Verify – verify that threats have been eliminated through follow-up audits

187. A client application needs to terminate a TCP communication session with a server. Place the termination process steps in the order that they will occur. (Not all options are used.)



Place the options in the following order:

step 1	client sends FIN
step 2	server sends ACK
step 3	server sends FIN
step 4	client sends ACK

Explanation: Topic 9.2.3

In order to terminate a TCP session, the client sends to the server a segment with the FIN flag set. The server acknowledges the client by sending a segment with the ACK flag set. The server sends a FIN to the client to terminate the server to client session. The client acknowledges the termination by sending a segment with the ACK flag set.

188. Match the attack vector with the description.



Category	Description
web	initiated through an email attachment
email	email
attrition	initiated from external storage
media	media
	uses brute force against devices or services
	attrition
	initiated from a website application
	web

Explanation: Topic 28.4.6

189. Match the security management function with the description.

Categories:	Options:
configuration management	the comprehensive analysis of impacts of attacks on core company assets and functioning
vulnerability management	the implementation of systems that track the location and configuration of networked devices and software across an enterprise
risk management	the security practice designed to proactively prevent the exploitation of IT vulnerabilities that exist within an organization
asset management	the inventory and control of hardware and software configurations of systems

Explanation: Topic 23.3

Place the options in the following order:

risk management	the comprehensive analysis of impacts of attacks on core company assets and functioning
configuration management	the inventory and control of hardware and software configurations of systems
asset management	the implementation of systems that track the location and configuration of networked devices and software across an enterprise
vulnerability management	the security practice designed to proactively prevent the exploitation of IT vulnerabilities that exist within an organization

190. What are two functions that are provided by the network layer? (Choose two.)

- directing data packets to destination hosts on other networks
- placing data on the network medium
- carrying data between processes that are running on source and destination hosts
- providing dedicated end-to-end connections
- providing end devices with a unique network identifier

Explanation: Topic 6.2.1

The network layer is primarily concerned with passing data from a source to a destination on another network. IP



destination. The network layer provides connectionless, best-effort delivery. Devices rely on higher layers to supply services to processes.

191. Match the phase in the NIST incident response life cycle to the action.

preparation	Document incident handling.
detection and analysis	post-incident activities
containment, eradication, and recovery	Conduct CSIRT response training.
post-incident activities	preparation
	Identify, analyze, and validate an incident.
	detection and analysis
	Implement procedures to contain the threat.
	containment, eradication, and recovery

Explanation: Topic 28.4.4

- Document incident handling. --> post-incident activities
- Conduct CSIRT response training. --> preparation
- Identify, analyze, and validate an incident. --> detection and analysis
- Implement procedures to contain the threat. --> containment, eradication, and recovery



192. Match the Linux CLI commands to their function.
(Not all options are used.)

rm

man

ls

cd

mkdir

copies files from source to destination

changes the current directory

cd

removes files

rm

creates a directory under the current directory

mkdir

displays the files inside a directory

ls

moves files to a different directory

displays the documentation for a specific command

man

Explanation: Topic 4.2.3



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